#12 · 3.4% of marks

# Ch 3 — Advanced Capital Budgeting Decisions

SM TYK theory Qs: **certainty equivalent approach · sensitivity analysis · project appraisal under inflation · steps in simulation analysis.** Numericals: inflation NPV, ENPV / σ / CV, RADR, CE, sensitivity, scenario, decision tree, replacement cycle (EAC), APV.

## ① Current trends — factors (SM 1)

*Intermediate assumed certain cash flows; in practice projects face (i) **inflation** (ii) **change in technology** (iii) **change in government policies**.*

## ② Inflation in capital budgeting (SM 1.1) — TYK Q3

→ *Inflation raises revenues and costs; net revenues after adjusting for inflation = net revenues in current terms. Two distortions:*

→ ***(1) Depreciation on historical cost**: after-tax CF = (R − C)(1 − T) + DT — (R − C)(1 − T) moves with inflation, **DT does not** → inflation **reduces the real rate of return**. SM eg: ₹80,000, 4 yrs, revenue 60,000, costs 20,000, tax 50% → CF 30,000 p.a.; with 10% inflation nominal CF 32,000 → 39,282 but **real** CF 29,091 → 26,830 (falls)*

→ ***(2) Cost of capital contains an inflation premium**: $R_N = R_R + P$ (SM's formula). If the rate has an inflation premium, reflect inflation in cash flows; no inflation → discount at the real rate*

***Illus 1**: outlay 40,000, real CF 15,000 × 4 yrs, cost of capital 12% incl. 10% inflation premium. Discounting real CF at 12% is **inconsistent**. Either (i) nominal CF = 15,000 × $1.10^t$ at 12% → NPV **₹17,369**, or (ii) real rate = 1.12 ÷ 1.10 − 1 = **1.8%** on real CF → **₹17,367** — same answer.*

***Different inflation for revenue and cost**: $NPV = \Sigma\ [\{R_t\ \Pi(1 + i_r) - C_t\ \Pi(1 + i_c)\}(1 - T) + D_tT] \div (1 + k)^t - I_0$. **Illus 2**: ₹8 lakh, revenue inflation 10/9/8/7%, cost 12/10/9/8% (compounded), tax 60%, dep 2 lakh (shield 1.2 lakh, not inflated), k 10% → NPV **₹1,06,744**.*

**⚠ TRAP**

*Inflate cumulatively: year-3 revenue = R × 1.10 × 1.09 × 1.08. The depreciation tax shield is **never** inflated.*

## ③ Technology change (SM 1.2)

***Why analyse**: alters production process · benefits (quality, delivery, flexibility) · may lower cost of capital · better cash inflows · may need extra capex · shorter product life cycle. **How**: (1) at decision time use sensitivity / scenario / simulation (2) after launch, revise estimates (3) if unviable, evaluate **abandonment** (4) adjust the discount rate. Review continuously.*

## 4 Government policies (SM 1.3)

- → Affect the firm directly or via suppliers, buyers, service providers; positive or negative; revise estimates, consider abandonment. Some change rarely (New Industrial Policy 1991), others within a year: **fiscal policy** (spending, taxation, borrowing) and **monetary policy** (OMO, reserve requirements, selective credit control, currency supply, repo; MPC meets at least **4 times a year**)
- → **Domestic**: (a) interest-rate changes affect cost of debt / YTM — little effect on long-term-funded capex but big on **working capital** (overdraft); small changes ignored, ~**100 bps** calls for revision (b) fiscal policy changes tax and depreciation rates → revise cash flows
- → **International**: (a) bank rate and money supply move exchange rates (b) tax on foreign income / **DTAA** changes

## 5 Risk in investment decisions (SM 2)

- → Decisions under (i) **certainty** (ii) **risk** — probabilities can be assigned (iii) **uncertainty** — they can't
- → **Risk** = variability of actual vs estimated returns; measured by σ and CV. Risk: distribution known; uncertainty: not known (terms used interchangeably)
- → **Why adjust for risk**: (1) opportunity cost — are returns proportionate to risk vs other options? (2) to know the real value of inflows — higher risk, higher premium and return

## 6 Internal & external risk factors (SM 3)

| Factor | SM meaning & example |
|---|---|
| **Project-specific** (internal) | Timely completion, estimation errors in resources & cash flows — nuclear vs hydel project |
| **Company-specific** (internal) | Rating downgrade, KMP changes, IPR / legal violations, labour disputes — two banks with different default exposure |
| **Industry-specific** | Regulatory restrictions, technology — leather, breweries |
| **Market** | Substitutes, demand, access to resources — coal supply to a thermal plant |
| **Competition** | Rival entry, product dynamism, change in tastes |
| **Economic conditions** | Monetary & fiscal changes (new taxes, cess), inflation, GDP, savings, disposable income |
| **International** | Trade / market-access restrictions, recession, bilateral agreements, political & geographical — limits on outsourcing jobs abroad |

## 7 Techniques of risk analysis (SM 4)

**Statistical**

Probability · variance / σ · coefficient of variation

**Conventional**

Risk-adjusted discount rate · certainty equivalent

**Others**

Sensitivity · scenario · simulation · decision tree

TYK THEORY Q1

## ⑧ Statistical techniques (SM 4.1)

**ENCF** = $\Sigma\ NCF_i \times P_i$ · **ENPV** = $\Sigma\ ENCF_t \div (1 + k)^t$ (incl. outflows) · $\sigma = \sqrt{\Sigma\ P_i(NCF_i - ENCF)^2}$ · **CV = σ ÷ expected return / cash flow**

CV = risk per unit of expected return — compare projects of different size; **lower CV preferred**. SM eg: 3,00,000 × 0.3 + 2,00,000 × 0.6 + 1,20,000 × 0.1 = **₹2,22,000**.

## ⑨ Risk-adjusted discount rate (SM 4.2.1)

→ Investors demand more from risky projects: rate = compensation for delaying consumption + inflation (= **risk-free**) + **risk premium** (depends on perceived risk & risk aversion). **RADR = $R_f$ + risk premium**

→ **Illus 7**: ₹100 lakh, CF 25/60/75/80/65 → at 7% NPV **144.34**; at RADR 14% NPV **99.79**

→ **Advantages**: (1) easy to understand (2) builds the premium into the discount rate. **Limitations**: (1) hard to find the premium / RADR (2) NPV yes, but **σ of the project can't be computed**

## ⑩ Certainty equivalent (SM 4.2.2) — TYK Q1

**CIMA**: expressing risky future cash flows as the certain cash flow the decision-maker would consider equivalent — **indifferent** between the risky amount and the (lower) riskless amount. Lets the decision-maker build in his **utility function**; a set of riskless cash flows replaces the original.

1. **Step 1**: $\alpha_t$ = certain CF ÷ risky (expected) CF; multiply each CF by $\alpha_t$. Coin toss: win ₹10,000 or nothing; indifferent at ₹3,000 certain → **CE ₹3,000, α = 0.3**
2. **Step 2**: discount at the **risk-free rate** — cost of capital would **double-count** risk
3. **Step 3**: normal capital budgeting; for **IRR compare with risk-free rate**, not the required return

α between 0 and 1; α = 1 → certain or risk-neutral; industry usually risk-averse. **NPV = $\Sigma\ \alpha_t\ NCF_t \div (1 + k_f)^t - 1$**. **Illus 8**: ₹45 lakh; 10/15/20/25 lakh × 0.90/0.85/0.82/0.78 at 5% → **₹5,34,570**.

**Advantages**

(1) simple to understand and apply (2) easily set for different risk levels in different years

**Disadvantages**

(1) no objective / mathematical way to estimate — subjective (2) reflects **management's** risk perception, ignores **shareholders'** → rarely used

**RADR vs CE**: CE is **superior** — it doesn't assume risk grows with time at a constant rate (each year's α fits its risk). But RADR is preferred in practice — easier to adjust a rate than specify a series of α.

# 11 Sensitivity analysis (SM 4.3.1) — TYK Q2

- → **CIMA**: modelling and risk-assessment procedure in which changes are made to significant variables to find their effect on the planned outcome; special attention to variables of special significance
- → Change **one variable at a time**; the more sensitive the NPV (IRR), the more **critical** the variable. **Steps**: (1) find variables influencing NPV / IRR (2) establish mathematical relationships (3) analyse effect of change in each
- → SM Illus 9: 2.5% adverse change in each variable → **selling price** has the maximum effect (NPV −24.82%)
- → **Advantages**: (1) identifies **critical issues** (2) **simple**. **Disadvantages**: (1) assumes variables are **independent** (2) **ignores probability** of changes

# 12 Scenario analysis (SM 4.3.2)

Extends sensitivity: brings in **probabilities** and changes **several variables at once**. Base (most likely) case → worst case (low sales & price, high variable cost) → best case; or mixed. Variability of outcomes across scenarios (eg boom / normal / recession) measures risk. **More complex** than sensitivity — all inputs change simultaneously.

# 13 Simulation — Monte Carlo (SM 4.3.3) — TYK Q4

Exact replica of the actual situation via a model; infinite calculations for possible outcomes and probabilities; ties together sensitivities and probability distributions; named after casino gambling (from first nuclear-bomb work); gives a **probability distribution of NPV**, not a point estimate.

1. **Model the project** — NPV vs **parameters** (set by decision-maker, constant across runs) and **exogenous variables** (stochastic, outside control)
2. Specify parameter values and probability distributions of exogenous variables
3. Select a value **at random** from each exogenous distribution
4. Compute NPV for those values
5. **Repeat** 3 & 4 many times
6. Plot the NPV distribution; compute **mean and σ**

**Advantages — "strength lies in variability"**

(1) predict bad market situations beforehand (2) handles many exogenous variables with any distribution, complex inter-relationships beyond analytical methods, forces explicit consideration of inter-dependencies

**Shortcomings (4)**

(1) hard to model and specify distributions (2) inherently **imprecise** — misleading when a tail matters (3) built by experts, decision-maker may not understand / use it (4) NPV at **risk-free** rate (to avoid pre-judging risk) → hard to interpret

# 14 Decision tree (SM 4.3.4)

- → Graphic display of the link between a present decision and future events, decisions and consequences — for **sequential** decisions. **Decision** (manager controls; acts rationally) vs **event** (no control; assign probabilities)
- → **Steps**: (1) define investment (launch local / national / international; extend factory vs outsource; dig for oil) (2) identify alternatives (3) draw the tree (4) evaluate — from the **last decision (right) backwards**; at each decision point keep highest NPV, truncate others; compute expected NPV at chance points
- → Rules: decision node = **rectangle**; chance node = **circle**; alternatives as straight lines; drawn **left to right**; number nodes; add values & probabilities

## 15 Replacement decisions (SM 5)

→ **Replace existing machine** (newer version or a different machine with the same function), mostly to maintain existing levels of operation. Method (app-written): incremental NPV — cost of new less sale of old, change in operating cash flows and depreciation tax shield, terminal values

→ **Optimum replacement cycle** (continuous replacement): NPVs of different cycle lengths aren't comparable → **EAC = PV of costs ÷ PVAF for the cycle**; choose the **lowest EAC**. SM taxi: EAC 1 yr 3,40,044 · 2 yr **3,15,331** · 3 yr 3,17,639 → replace every **2 years**. ICAI MTP machine: 4-year cycle lowest at ₹58,656

## 16 Adjusted Present Value (SM 6)

**APV = Base-case NPV + PV of tax benefit on interest**. Base case at the **unlevered** cost of equity (all-equity firm; unlevered $K_e$ from Ch 13). **Separates investment and financing decisions**; credits the project for the debt used to finance it.

## 17 Correlation — how Ch 3 actually connects

| Ch 3 idea | The actual link (app-written cross-reference; the Ch 3 side is SM) |
|---|---|
| Real vs nominal (1.12 ÷ 1.10 − 1) | **Ch 5**: equity is valued on **nominal** CF at nominal rates; **Ch 11**: PPP uses the same inflation ratio to forecast exchange rates. |
| RADR = $R_f$ + premium | **Ch 6**: CAPM gives the premium objectively — $\beta(R_m - R_f)$; **Ch 11**: SM says don't load political / FX risk into the rate — use certainty equivalents. |
| Certainty equivalent at $R_f$ | **Ch 9**: risk-neutral valuation also discounts at $R_f$ after adjusting the cash flows / probabilities — same "adjust numerator, not denominator" logic. |
| σ and CV of NPV | **Ch 6**: σ and CV of returns; **Ch 2**: σ × 2 = VaR. |
| Abandonment, decision tree | **Ch 9**: real options (abandon, expand, wait) give value that static NPV misses; decision trees value them. |
| APV | **Ch 11**: SM's APV for foreign projects adds interest subsidies; **Ch 13**: unlevered $K_e$ via asset beta. |
| Replacement cycle / EAC | **FM**: EAC from Intermediate; equal-life comparison logic as in lease vs buy. |

# ③ Advanced Capital Budgeting Decisions — what ICAI asked (Sep 2025, Jan 2026, May 2026)

***12 marks*** *set across the last three attempts (3.4% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

| Attempt | Question | Marks | Concept tested |
|---|---|---|---|
| Sep 2025 | MCQ CS-I 1-3 | 6 | Sensitivity analysis of NPV: % change in NPV when initial cost, annual cash inflow, or cost of capital is varied adversely by 10% (MCQ) |
| Jan 2026 | MCQ CS-V 13-15 | 6 | Choice between a sure product and an uncertain one: profit from contribution minus fixed cost, and expected value of profit using a probability distribution (MCQ) |

*What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.*

# Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

## Theory Q1

1. Write short note on Certainty Equivalent Approach.

### ICAI answer

**1.** This approach recognizes risk in capital budgeting analysis by adjusting estimated cash flows and employs risk free rate to discount the adjusted cash-flows. Under this method, the expected cash flows of the project are converted to equivalent riskless amounts. The greater the risk of an expected cash flow, the smaller the certainty equivalent values for receipts and longer the CE value for payment. This approach is superior to the risk adjusted discounted approach as it can measure risk more accurately.

This is yet another approach for dealing with risk in capital budgeting to reduce the forecasts of cash flows to some conservative levels. In certainty Equivalent approach we incorporate risk to adjust the cash flows of a proposal so as to reflect the risk element. The certainty Equivalent approach adjusts future cash flows rather than discount rates. This approach explicitly recognizes risk, but the procedure for reducing the forecasts of cash flows is implicit and likely to be inconsistent from one investment to another.

## Theory Q2

2. What is the sensitivity analysis in Capital Budgeting?

### ICAI answer

**2.** Sensitivity analysis is used in Capital budgeting for more precisely measuring the risk. It helps in assessing information as to how sensitive are the estimated parameters of the project such as cash flows, discount rate, and the project life to the estimation errors. Future being always uncertain and estimations are always subject to error, sensitivity analysis takes care of estimation errors by using a number of possible outcomes in evaluating a project. The

methodology adopted in sensitivity analysis is to evaluate a project by using a number of estimated cash flows so as to provide to the decision maker an insight into the variability of outcome. Thus, it is a technique of risk analysis which studies the responsiveness of a criterion of merit like NPV or IRR to variation in underlying factors like selling price, quantity sold, returns from an investment etc.

Sensitivity analysis answers questions like,

(i) What happens to the present value (or some other criterion of merit) if flows are, say ₹ 50,000 than the expected ₹ 80,000?

(ii) What will happen to NPV if the economic life of the project is only 3 years rather than expected 5 years?

Therefore, wherever there is an uncertainty, of whatever type, the sensitivity analysis plays a crucial role. However, it should not be viewed as the method to remove the risk or uncertainty, it is only a tool to analyse and measure the risk and uncertainty. In terms of capital budgeting the possible cash flows are based on three assumptions:

(a) Cash flows may be worst (pessimistic)

(b) Cash flows may be most likely.

(c) Cash flows may be most optimistic.

Sensitivity analysis involves three steps

(1) Identification of all those variables having an influence on the project's NPV or IRR.

(2) Definition of the underlying quantitative relationship among the variables.

(3) Analysis of the impact of the changes in each of the variables on the NPV of the project.

The decision maker, in sensitivity analysis always asks himself the question – what if?

## Theory Q3

3. Write a note on project appraisal under inflationary conditions.

## ICAI answer

**3.** Project Appraisal normally involves feasibility evaluation from technical, commercial, economic and financial aspects. It is generally an exercise in measurement and analysis of cash flows expected to occur over the life of the project. The project cash outflows usually occur initially and inflows come in the future.

During inflationary conditions, the project cost increases on all heads viz. labour, raw material, fixed assets such as equipments, plant and machinery, building material, remuneration of technicians and managerial personnel etc. Beside this, inflationary conditions

erode purchasing power of consumers and affect the demand pattern. Thus, not only cost of production but also the projected statement of profitability and cash flows are affected by the change in demand pattern. Even financial institutions and banks may revise their lending rates resulting in escalation in financing cost during inflationary conditions. Under such circumstances, project appraisal has to be done generally keeping in view the following guidelines which are usually followed by government agencies, banks and financial institutions.

(i) It is always advisable to make provisions for cost escalation on all heads of cost, keeping in view the rate of inflation during likely period of delay in project implementation.

(ii) The various sources of finance should be carefully scrutinized with reference to probable revision in the rate of interest by the lenders and the revision which could be affected in the interest-bearing securities to be issued. All these factors will push up the cost of funds for the organization.

(iii) Adjustments should be made in profitability and cash flow projections to take care of the inflationary pressures affecting future projections.

(iv) It is also advisable to examine the financial viability of the project at the revised rates and assess the same with reference to economic justification of the project. The appropriate measure for this aspect is the economic rate of return for the project which will equate the present value of capital expenditures to net cash flows over the life of the projects. The rate of return should be acceptable which also accommodates the rate of inflation per annum.

(v) In an inflationary situation, projects having early payback periods should be preferred because projects with long payback period are riskier.

Under conditions of inflation, the project cost estimates that are relevant for a future date will suffer escalation. Inflationary conditions will tend to initiate the measurement of future cash flows. Either of the following two approaches may be used while appraising projects under such conditions:

(a) Adjust each year's cash flows to an inflation index, recognizing selling price increases and cost increases annually; or

(b) Adjust the 'Acceptance Rate' (cut-off) suitably retaining cash flow projections at current price levels.

An example of approach (ii) above can be as follows:

| | | |
|---|---|---|
| Normal Acceptance Rate | : | 15.0% |
| Expected Annual Inflation | : | 5.0% |
| Adjusted Discount Rate | : | 15.0 × 1.05 or 15.75% |

It must be noted that measurement of inflation has no standard approach nor is easy. This makes the job of appraisal a difficult one under such conditions.

## Theory Q4

4. Explain the steps involved in Simulation Analysis.

*ICAI answer: "Please refer paragraph 4.3.3" – SM paragraph 4.3.3 reproduced*

### 4.3.3 Simulation Analysis (Monte Carlo)

Simulation is the exact replica of the actual situation. To simulate an actual situation, a model shall be prepared. The simulation Analysis is a technique, in which infinite calculations are made to obtain the possible outcomes and probabilities for any given action.

Monte Carlo simulation ties together sensitivities and probability distributions. The method came out of the work of first nuclear bomb and was so named because it was based on mathematics of Casino gambling. Fundamental appeal of this analysis is that it provides decision makers with a probability distribution of NPVs rather than a single point estimates of the expected NPV.

This analysis starts with carrying out a simulation exercise to model the investment project. It involves identifying the key factors affecting the project and their inter relationships. It involves modelling of cash flows to reveal the key factors influencing both cash receipt and payments and their inter relationship.

This analysis specifies a range for a probability distribution of potential outcomes for each of model's assumptions.

**4.3.3.1 Steps for Simulation Analysis:**

1. Modelling the project: The model shows the relationship of NPV with parameters and exogenous variables. (Parameters are input variables specified by decision maker and held constant over all simulation runs. Exogenous variables are input variables, which are stochastic in nature and outside the control of the decision maker).
2. Specify values of parameters and probability distributions of exogenous variables.
3. Select a value at random from probability distribution of each of the exogenous variables.
4. Determine NPV corresponding to the randomly generated value of exogenous variables and pre-specified parameter variables.
5. Repeat steps (3) & (4) a large number of times to get a large number of simulated NPVs.
6. Plot probability distribution of NPVs and compute a mean and Standard Deviation of returns to gauge the project's level of risk.

**Example:** Uncertainty associated with two aspects of the project: Annual Net Cash Flow & Life of the project. NPV model for the project is

$$\sum_{t=1}^{n} [CF_t/(1+i)^t] - I$$

Where i→ Risk free interest rate, I→ initial investment are parameters, CF = Annual Cash Flow

With i = 10%, I = ₹ 1,30,000, $CF_t$ & n stochastic exogenous variables with the following distribution will be as under:

| Annual Cash Flow | | Project Life | |
|---|---|---|---|
| Value (₹) | Probability | Value (Year) | Probability |
| 10,000 | 0.02 | 3 | 0.05 |
| 15,000 | 0.03 | 4 | 0.10 |
| 20,000 | 0.15 | 5 | 0.30 |
| 25,000 | 0.15 | 6 | 0.25 |
| 30,000 | 0.30 | 7 | 0.15 |
| 35,000 | 0.20 | 8 | 0.10 |
| 40,000 | 0.15 | 9 | 0.03 |
| | | 10 | 0.02 |

Ten manual simulation runs are performed for the project. To perform this operation, values are generated at random for the two exogenous variables viz., Annual Cash Flow and Project Life. For this purpose, we take following steps

(1) set up correspondence between values of exogenous variables and random numbers

(2) choose some random number generating device.

Correspondence between Values of Exogenous Variables and two Digit Random Numbers:

| Annual Cash Flow | | | | Project Life | | | |
|---|---|---|---|---|---|---|---|
| Value (₹) | Probability | Cumulative Probability | Two Digit Random No. | Value (Year) | Probability | Cumulative Probability | Two Digit Random No. |
| 10,000 | 0.02 | 0.02 | 00 – 01 | 3 | 0.05 | 0.05 | 00 – 04 |
| 15,000 | 0.03 | 0.05 | 02 – 04 | 4 | 0.10 | 0.15 | 05 – 14 |
| 20,000 | 0.15 | 0.20 | 05 – 19 | 5 | 0.30 | 0.45 | 15 – 44 |
| 25,000 | 0.15 | 0.35 | 20 – 34 | 6 | 0.25 | 0.70 | 45 – 69 |
| 30,000 | 0.30 | 0.65 | 35 – 64 | 7 | 0.15 | 0.85 | 70 – 84 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 35,000 | 0.20 | 0.85 | 65 – 84 | 8 | 0.10 | 0.95 | 85 – 94 |
| 40,000 | 0.15 | 1.00 | 85 - 99 | 9 | 0.03 | 0.98 | 95 – 97 |
| | | | | 10 | 0.02 | 1.00 | 98 - 99 |

**Random Number**

| | | | | |
|---|---|---|---|---|
| 53479 | 81115 | 98036 | 12217 | 59526 |
| 97344 | 70328 | 58116 | 91964 | 26240 |
| 66023 | 38277 | 74523 | 71118 | 84892 |
| 99776 | 75723 | 03172 | 43112 | 83086 |
| 30176 | 48979 | 92153 | 38416 | 42436 |
| 81874 | 83339 | 14988 | 99937 | 13213 |
| 19839 | 90630 | 71863 | 95053 | 55532 |
| 09337 | 33435 | 53869 | 52769 | 18801 |
| 31151 | 58295 | 40823 | 41330 | 21093 |
| 67619 | 52515 | 03037 | 81699 | 17106 |

For random numbers, we can begin from any-where taking at random from the table and read any pair of adjacent columns, column/row wise. For the first simulation run we need two digit random numbers (1) For Annual Cash Flow (2) For Project Life. The numbers are 53 & 97 and corresponding value of Annual Cash Flow and Project Life are ₹ 3,000 and 9 years respectively.

**Simulation Results**

| Annual Cash Flow | | | Project Life | | | |
|---|---|---|---|---|---|---|
| Run | Random No. | Corres. Value of Annual Cash Flow (1) | Random No. | Corres. Value of Project Life | PVAF @ 10% (2) | NPV (1)x(2) – 1,30,000 |
| 1 | 53 | 30,000 | 97 | 9 | 5.759 | 42,770 |
| 2 | 66 | 35,000 | 99 | 10 | 6.145 | 85,075 |
| 3 | 30 | 25,000 | 81 | 7 | 4.868 | (8,300) |
| 4 | 19 | 20,000 | 09 | 4 | 3.170 | (66,600) |
| 5 | 31 | 25,000 | 67 | 6 | 4.355 | (21,125) |
| 6 | 81 | 35,000 | 70 | 7 | 4.868 | 40,380 |
| 7 | 38 | 30,000 | 75 | 7 | 4.868 | 16,040 |
| 8 | 48 | 30,000 | 83 | 7 | 4.868 | 16,040 |
| 9 | 90 | 40,000 | 33 | 5 | 3.791 | 21,640 |
| 10 | 58 | 30,000 | 52 | 6 | 4.355 | 650 |

(1) We can predict all type of bad market situation beforehand.

(2) Handle problems characterised by:

(a) numerous exogenous variables following any kind of distribution.

(b) complex inter-relationships among parameters, exogenous variables and endogenous variables. Such problems defy capabilities of analytical methods.

(c) compels decision maker to explicitly consider the inter-dependencies and uncertainties featuring the project.

**4.3.3.3 Shortcomings**

(1) Difficult to model the project and specify probability distribution of exogenous variables.

(2) Simulation is inherently imprecise. Provides rough approximation of probability distribution of NPV Due to its imprecision, simulation probability distribution may be misleading when a tail of distribution is critical.

(3) Realistic simulation model being likely to be complex would probably be constructed by management expert and not by the decision maker. Decision maker lacking understanding of the model may not use it.

(4) Determine NPV in simulation run, risk free discount rate is used. It is done to avoid pre-judging risk, which is reflected in the dispersion of the distribution of N.P.V. This derived measure of NPV takes a different meaning from its original value, and, therefore, is difficult to interpret.

*ICAI answer: "Please refer paragraph 4.3.3" – SM paragraph 4.3.3 reproduced*

### 4.3.3 Simulation Analysis (Monte Carlo)

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| Annual Cash Flow | | | | Project Life | | | |
|---|---|---|---|---|---|---|---|
| Value (₹) | Probability | Cumulative Probability | Two Digit Random No. | Value (Year) | Probability | Cumulative Probability | Two Digit Random No. |
| 10,000 | 0.02 | 0.02 | 00 – 01 | 3 | 0.05 | 0.05 | 00 – 04 |
| 15,000 | 0.03 | 0.05 | 02 – 04 | 4 | 0.10 | 0.15 | 05 – 14 |
| 20,000 | 0.15 | 0.20 | 05 – 19 | 5 | 0.30 | 0.45 | 15 – 44 |
| 25,000 | 0.15 | 0.35 | 20 – 34 | 6 | 0.25 | 0.70 | 45 – 69 |
| 30,000 | 0.30 | 0.65 | 35 – 64 | 7 | 0.15 | 0.85 | 70 – 84 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 35,000 | 0.20 | 0.85 | 65 – 84 | 8 | 0.10 | 0.95 | 85 – 94 |
| 40,000 | 0.15 | 1.00 | 85 - 99 | 9 | 0.03 | 0.98 | 95 – 97 |
| | | | | 10 | 0.02 | 1.00 | 98 - 99 |

**Random Number**

| | | | | |
|---|---|---|---|---|
| 53479 | 81115 | 98036 | 12217 | 59526 |
| 97344 | 70328 | 58116 | 91964 | 26240 |
| 66023 | 38277 | 74523 | 71118 | 84892 |
| 99776 | 75723 | 03172 | 43112 | 83086 |
| 30176 | 48979 | 92153 | 38416 | 42436 |
| 81874 | 83339 | 14988 | 99937 | 13213 |
| 19839 | 90630 | 71863 | 95053 | 55532 |
| 09337 | 33435 | 53869 | 52769 | 18801 |
| 31151 | 58295 | 40823 | 41330 | 21093 |
| 67619 | 52515 | 03037 | 81699 | 17106 |

For random numbers, we can begin from any-where taking at random from the table and read any pair of adjacent columns, column/row wise. For the first simulation run we need two digit random numbers (1) For Annual Cash Flow (2) For Project Life. The numbers are 53 & 97 and corresponding value of Annual Cash Flow and Project Life are ₹ 3,000 and 9 years respectively.

**Simulation Results**

| Annual Cash Flow | | | Project Life | | | |
|---|---|---|---|---|---|---|
| Run | Random No. | Corres. Value of Annual Cash Flow (1) | Random No. | Corres. Value of Project Life | PVAF @ 10% (2) | NPV (1)x(2) – 1,30,000 |
| 1 | 53 | 30,000 | 97 | 9 | 5.759 | 42,770 |
| 2 | 66 | 35,000 | 99 | 10 | 6.145 | 85,075 |
| 3 | 30 | 25,000 | 81 | 7 | 4.868 | (8,300) |
| 4 | 19 | 20,000 | 09 | 4 | 3.170 | (66,600) |
| 5 | 31 | 25,000 | 67 | 6 | 4.355 | (21,125) |
| 6 | 81 | 35,000 | 70 | 7 | 4.868 | 40,380 |
| 7 | 38 | 30,000 | 75 | 7 | 4.868 | 16,040 |
| 8 | 48 | 30,000 | 83 | 7 | 4.868 | 16,040 |
| 9 | 90 | 40,000 | 33 | 5 | 3.791 | 21,640 |
| 10 | 58 | 30,000 | 52 | 6 | 4.355 | 650 |

(1) We can predict all type of bad market situation beforehand.

(2) Handle problems characterised by:

(a) numerous exogenous variables following any kind of distribution.

(b) complex inter-relationships among parameters, exogenous variables and endogenous variables. Such problems defy capabilities of analytical methods.

(c) compels decision maker to explicitly consider the inter-dependencies and uncertainties featuring the project.

**4.3.3.3 Shortcomings**

(1) Difficult to model the project and specify probability distribution of exogenous variables.

(2) Simulation is inherently imprecise. Provides rough approximation of probability distribution of NPV Due to its imprecision, simulation probability distribution may be misleading when a tail of distribution is critical.

(3) Realistic simulation model being likely to be complex would probably be constructed by management expert and not by the decision maker. Decision maker lacking understanding of the model may not use it.

(4) Determine NPV in simulation run, risk free discount rate is used. It is done to avoid pre-judging risk, which is reflected in the dispersion of the distribution of N.P.V. This derived measure of NPV takes a different meaning from its original value, and, therefore, is difficult to interpret.

# ① SM illustrations with ICAI solutions

## I. Current Trends In Capital Budgeting

### SM Ch 3 · Illustration 1 (under 1.1 Impact of Inflation on Capital Budgeting Decisions)

**Illustration 1**

Determine NPV of the project with the following information:

| | |
|---|---|
| Initial Outlay of project | ₹ 40,000 |
| Annual Cash Flow from the Project (Without inflation) | ₹ 15,000 |
| Useful life | 4 years |
| Cost of Capital (Including inflation premium of 10%) | 12% |

### Solution — ICAI

**Solution**

Annual Cash Flow of project is ₹ 15,000.

It would be inconsistent to discount these real cash flows at 12% (nominal rate of return).

There are two alternatives:

(i) Either to restate the cash flow in nominal term and discount at 12% or

(ii) Restate the discount rate in real terms and use this to discount the real cash flows.

**NPV using (i) approach**

Since inflation rate is 10% a year, real cash flows may be stated in nominal cash flows as follows:

Nominal Cash Flow = (1 + Inflation Rate) Real Cash Flows

| Year | Real Cash Flows | Nominal Cash flows |
|---|---|---|
| 1 | 15000 | 15,000 × 1.10 = 16,500 |
| 2 | 15,000 | 15,000 × $(1.10)^2$ = 18,150 |
| 3 | 15,000 | 15,000 × $(1.10.)^3$ = 19,965 |
| 4 | 15,000 | 15,000 × $(1.10)^4$ = 21,962 |

NPV using nominal discounting rate 12%

$$\frac{16,500}{(1.12)}+\frac{18,150}{(1.12)^2}+\frac{19,965}{(1.12)^3}+\frac{21,962}{(1.12)^4}-40,000$$

= ₹ 14,732 + ₹ 14,469 + ₹ 14,211+ ₹ 13,957 – ₹ 40,000

= ₹ 17,369 (Approx)

**NPV using (ii) approach**

To compute NPV using (ii) approach, we shall need real discount rate, which shall be computed as follows:

$$\text{Real Discount Rate}=\frac{1+\text{Nominal Discount Rate}}{1+\text{Inflation Rate}}-1$$

$$\text{Real Discount Rate}=\frac{1+0.12}{1+0.10}-1=0.0182 \text{ i.e. } 1.8\%.$$

$$NPV = \sum_{t=1}^{n} cf_t - I_o$$

Where t = Time Period

$cf_t$ = Annual Cash Flow

$I_o$ = Initial Outlay

Accordingly NPV of the project

$$\frac{15,000}{(1.0182)} + \frac{15,000}{(1.0182)^2} + \frac{15,000}{(1.0182)^3} + \frac{15,000}{(1.0182)^4} - 40,000$$

= ₹ 14,732 + ₹ 14,469 + ₹ 14,210+ ₹ 13,956 – ₹ 40,000

= ₹ 57,367 – ₹ 40,000 = ₹17,367(Approx)

NPV based on consideration that inflation rate for revenue and cost are different shall be computed as follows:

$$N.P.V. = {}^{n}\Sigma_{t=1} [\{R_t (1+i_r) - C_t{}^{t}\Sigma_{r=1}(1+i_c)\} (1-T) + D_t T] / (1+k)^t - I_0$$

$R_t$→ revenues for the year 't' with no inflation.

$i_r$→ annual inflation rate in revenues for 'r th ' year.

$C_t$→ costs for year 't' with no inflation.

$i_c$→ annual inflation rate of costs for year 'r'.

T → tax rate.

$D_t$→ depreciation charge for year 't'.

$I_0$→ initial outlay.

k → cost of capital (with inflation premium).

## SM Ch 3 · Illustration 2 (under 1.1 Impact of Inflation on Capital Budgeting Decisions)

### Illustration 2

XYZ Ltd. requires ₹ 8,00,000 for a project. Useful life of project - 4 years. Salvage value - Nil. Depreciation Charge ₹ 2,00,000 p.a. Expected revenues & costs (excluding depreciation) ignoring inflation.

| Year | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Revenues | ₹ 6,00,000 | ₹ 7,00,000 | ₹ 8,00,000 | ₹ 8,00,000 |
| Costs | ₹ 3,00,000 | ₹ 4,00,000 | ₹ 4,00,000 | ₹ 4,00,000 |

Applicable Tax Rate is 60% and cost of capital is 10% (including inflation premium).

Calculate NPV of the project if inflation rates for revenues & costs are as follows:

| Year | Revenues | Costs |
|---|---|---|
| 1 | 10% | 12% |
| 2 | 9% | 10% |
| 3 | 8% | 9% |
| 4 | 7% | 8% |

## Solution — ICAI

**Solution**

**Computation of Annual Cash Flow**

(i) Inflation adjusted Revenues

| Year | Revenues (₹) | Revenues (Inflation Adjusted) (₹) |
|---|---|---|
| 1 | 6,00,000 | 6,00,000(1.10) = 6,60,000 |
| 2 | 7,00,000 | 7,00,000(1.10)(1.09) = 8,39,300 |
| 3 | 8,00,000 | 8,00,000(1.10)(1.09)(1.08) = 10,35,936 |
| 4 | 8,00,000 | 8,00,000(1.10)(1.09)(1.08)(1.07) = 11,08,452 |

(ii) Inflation adjusted Costs

| Year | Revenues (₹) | Revenues (Inflation Adjusted) (₹) |
|---|---|---|
| 1 | 3,00,000 | 3,00,000(1.12) = 3,36,000 |
| 2 | 4,00,000 | 4,00,000(1.12)(1.10) = 4,92,800 |
| 3 | 4,00,000 | 4,00,000(1.12)(1.10)(1.09) = 5,37,152 |
| 4 | 4,00,000 | 4,00,000(1.12)(1.10)(1.09)(1.08) = 5,80,124 |

(iii) Tax Benefit on Depreciation = ₹ 2,00,000 x 0.60 = ₹ 1,20,000

(iv) Net Profit after Tax

| Year | Revenues (Inflation Adjusted) (₹)(1) | Costs (Inflation Adjusted) (₹)(2) | Net Profit (₹) (3) = (1) - (2) | Tax (₹) (4) = 60% of (3) | Net after Profit (₹) (3) - (4) |
|---|---|---|---|---|---|
| 1 | 6,60,000 | 3,36,000 | 3,24,000 | 1,94,400 | 1,29,600 |
| 2 | 8,39,300 | 4,92,800 | 3,46,500 | 2,07,900 | 1,38,600 |
| 3 | 10,35,936 | 5,37,152 | 4,98,784 | 2,99,270 | 1,99,514 |
| 4 | 11,08,452 | 5,80,124 | 5,28,328 | 3,16,997 | 2,11,331 |

(iv) Present Value of Cash Inflows

| Year | Net after Profit (₹) | Tax Benefit on Depreciation (₹) | Cash Inflow (₹) | PVF@ 10% | PV (₹) |
|---|---|---|---|---|---|
| 1 | 1,29,600 | 1,20,000 | 2,49,600 | 0.909 | 2,26,886 |
| 2 | 1,38,600 | 1,20,000 | 2,58,600 | 0.826 | 2,13,604 |
| 3 | 1,99,514 | 1,20,000 | 3,19,514 | 0.751 | 2,39,955 |
| 4 | 2,11,331 | 1,20,000 | 3,31,331 | 0.683 | 2,26,299 |
| | | | | | 9,06,744 |

NPV = ₹ 9,06,744 – ₹ 8,00,000 = ₹ 1,06,744

# 4. Methods Of Incorporating Risk In Capital

## SM Ch 3 · Illustration 3 (under 4.1.1 Probability)

### Illustration 3

Possible net cash flows of Projects A and B at the end of first year and their probabilities are given below. Discount rate is 10 per cent. For both the projects, initial investment is ₹ 10,000. Calculate the expected net present value for each project. State which project is preferable?

| Possible Event | Project A | | Project B | |
|---|---|---|---|---|
| | Cash Flow (₹) | Probability | Cash Flow (₹) | Probability |
| A | 8,000 | 0.10 | 24,000 | 0.10 |
| B | 10,000 | 0.20 | 20,000 | 0.15 |
| C | 12,000 | 0.40 | 16,000 | 0.50 |
| D | 14,000 | 0.20 | 12,000 | 0.15 |
| E | 16,000 | 0.10 | 8 ,000 | 0.10 |

## Solution — ICAI

### Solution

**Calculation of Expected Value for Project A and Project B**

| | Project A | | | Project B | | |
|---|---|---|---|---|---|---|
| Possible Event | Cash Flow (₹) | Probability | Expected Value (₹) | Cash Flow (₹) | Probability | Expected Value (₹) |
| A | 8,000 | 0.10 | 800 | 24,000 | 0.10 | 2,400 |
| B | 10,000 | 0.20 | 2,000 | 20,000 | 0.15 | 3,000 |
| C | 12,000 | 0.40 | 4,800 | 16,000 | 0.50 | 8,000 |
| D | 14,000 | 0.20 | 2,800 | 12,000 | 0.15 | 1,800 |
| E | 16,000 | 0.10 | 1,600 | 8,000 | 0.10 | 800 |
| ENCF | | | 12,000 | | | 16,000 |

The Net Present Value for Project A is (0.909 × ₹ 12,000 – ₹ 10,000) = ₹ 908

The Net Present Value for Project B is (0.909 × ₹ 16,000 – ₹ 10,000) = ₹ 4,544.

**(b) Expected Net Present Value- Multiple period**

Let us understand the calculation of Expected Net Present Value (ENPV) for multiple periods through an illustration as follows:

## SM Ch 3 · Illustration 4 (under 4.1.1 Probability)

### Illustration 4

Probabilities for net cash flows for 3 years of a project are as follows:

| Year 1 | | Year 2 | | Year 3 | |
|---|---|---|---|---|---|
| Cash Flow (₹) | Probability | Cash Flow (₹) | Probability | Cash Flow (₹) | Probability |
| 2,000 | 0.1 | 2,000 | 0.2 | 2,000 | 0.3 |
| 4,000 | 0.2 | 4,000 | 0.3 | 4,000 | 0.4 |
| 6,000 | 0.3 | 6,000 | 0.4 | 6,000 | 0.2 |
| 8,000 | 0.4 | 8,000 | 0.1 | 8,000 | 0.1 |

Calculate the expected net present value of the project using 10 per cent discount rate if the Initial Investment of the project is ₹ 10,000.

## Solution — ICAI

### Solution

Calculation of Expected Value

| Year 1 | | | Year 2 | | | Year 3 | | |
|---|---|---|---|---|---|---|---|---|
| Cash Flow (₹) | Prob. | Expected Value (₹) | Cash Flow (₹) | Prob. | Expected Value (₹) | Cash Flow (₹) | Prob. | Expected Value (₹) |
| 2,000 | 0.1 | 200 | 2,000 | 0.2 | 400 | 2,000 | 0.3 | 600 |
| 4,000 | 0.2 | 800 | 4,000 | 0.3 | 1200 | 4,000 | 0.4 | 1,600 |
| 6,000 | 0.3 | 1,800 | 6,000 | 0.4 | 2400 | 6,000 | 0.2 | 1,200 |
| 8,000 | 0.4 | 3,200 | 8,000 | 0.1 | 800 | 8,000 | 0.1 | 800 |
| ENCF | | 6,000 | | | 4,800 | | | 4,200 |

The present value of the expected value of cash flow at 10 per cent discount rate has been determined as follows:

Present Value of cash flow $= \frac{ENCF_1}{(1+k)^1} + \frac{ENCF_2}{(1+k)^2} + \frac{ENCF_3}{(1+k)^3}$

$= \frac{6,000}{(1.1)} + \frac{4,800}{(1.1)^2} + \frac{4,200}{(1.1)^3}$

$= (6,000 \times 0.909) + (4,800 \times 0.826) + (4,200 \times 0.751)$

$= ₹ 12,573$

Expected Net Present value = Present Value of cash flow - Initial Investment

= ₹ 12,573 – ₹ 10,000 = ₹ 2,573.

## SM Ch 3 · Illustration 5 (under 4.1.3 Standard Deviation)

### Illustration 5

Calculate Variance and Standard Deviation of Project A and Project B on the basis of following information:

| Possible Event | Project A | | Project B | |
|---|---|---|---|---|
| | Cash Flow (₹) | Probability | Cash Flow (₹) | Probability |
| A | 8,000 | 0.10 | 24,000 | 0.10 |
| B | 10,000 | 0.20 | 20,000 | 0.15 |
| C | 12,000 | 0.40 | 16,000 | 0.50 |
| D | 14,000 | 0.20 | 12,000 | 0.15 |
| E | 16,000 | 0.10 | 8,000 | 0.10 |

## Solution — ICAI

### Solution

Calculation of Expected Value for Project A and Project B

| | Project A | | | Project B | | |
|---|---|---|---|---|---|---|
| Possible Event | Cash Flow (₹) | Probability | Expected Value (₹) | Cash Flow (₹) | Probability | Expected Value (₹) |
| A | 8,000 | 0.10 | 800 | 24,000 | 0.10 | 2,400 |
| B | 10,000 | 0.20 | 2,000 | 20,000 | 0.15 | 3,000 |
| C | 12,000 | 0.40 | 4,800 | 16,000 | 0.50 | 8,000 |
| D | 14,000 | 0.20 | 2,800 | 12,000 | 0.15 | 1,800 |
| E | 16,000 | 0.10 | 1,600 | 8,000 | 0.10 | 800 |
| ENCF | | | 12,000 | | | 16,000 |

**Project A:**

Variance ($\sigma 2$) = $(8,000 - 12,000)2 \times (0.1) + (10,000 - 12,000)2 \times (0.2) + (12,000 - 12000)2 \times (0.4) + (14,000 - 12,000)2 \times (0.2) + (16000 - 12,000)2 \times (0.1)$

= 16,00,000 + 8,00,000 + 0 + 8,00,000 + 16,00,000 = 48,00,000

Standard Deviation ($\sigma$) = $\sqrt{\text{Variance}(\sigma^2)}$ = $\sqrt{48,00,000}$ = 2,190.90

**Project B:**

Variance($\sigma 2$) = $(24,000 - 16,000)2 \times (0.1) + (20,000 - 16,000)2 \times (0.15) + (16,000 - 16,000)2 \times (0.5) + (12,000 - 16,000)2 \times (0.15) + (8,000 - 16,000)2 \times (0.1)$

= 64,00,000 + 24,00,000 + 0 + 24,00,000 + 64,00,000 = 1,76,00,000

Standard Deviation ($\sigma$) = $\sqrt{\text{Variance}(\sigma^2)}$ = $\sqrt{1,76,00,000}$ = 4195.23

## SM Ch 3 · Illustration 6 (under 4.1.4 The Coefficient of Variation)

### Illustration 6

Calculate Coefficient of Variation of Project A and Project B based on the following information:

| Possible Event | Project A | | Project B | |
|---|---|---|---|---|
| | Cash Flow (₹) | Probability | Cash Flow (₹) | Probability |
| A | 10000 | 0.10 | 26,000 | 0.10 |
| B | 12,000 | 0.20 | 22,000 | 0.15 |
| C | 14,000 | 0.40 | 18,000 | 0.50 |
| D | 16,000 | 0.20 | 14,000 | 0.15 |
| E | 18,000 | 0.10 | 10,000 | 0.10 |

## Solution — ICAI

### Solution

**Calculation of Expected Value for Project A and Project B**

| | Project A | | | Project B | | |
|---|---|---|---|---|---|---|
| Possible Event | Cash Flow (₹) | Probability | Expected Value (₹) | Cash Flow (₹) | Probability | Expected Value (₹) |
| A | 10,000 | 0.10 | 1,000 | 26,000 | 0.10 | 2,600 |
| B | 12,000 | 0.20 | 2,400 | 22,000 | 0.15 | 3,300 |
| C | 14,000 | 0.40 | 5,600 | 18,000 | 0.50 | 9,000 |
| D | 16,000 | 0.20 | 3,200 | 14,000 | 0.15 | 2,100 |
| E | 18,000 | 0.10 | 1,800 | 10,000 | 0.10 | 1,000 |
| ENCF | | | 14,000 | | | 18,000 |

**Project A**

Variance ($\sigma^2$) = $(10,000 - 14,000)^2 \times (0.1) + (12,000 - 14,000)^2 \times (0.2) + (14,000 - 14000)^2 \times (0.4) + (16,000 - 14,000)^2 \times (0.2) + (18000 - 14,000)^2 \times (0.1)$

= 16,00,000 + 8,00,000 + 0 + 8,00,000 + 16,00,000 = 48,00,000

Standard Deviation ($\sigma$) = $\sqrt{\text{Variance}(\sigma^2)} = \sqrt{48,00,000} = 2,190.90$

**Project B:**

Variance($\sigma^2$) = $(26,000 - 18,000)^2 \times (0.1) + (22,000 - 18,000)^2 \times (0.15) + (18,000 - 18,000)^2 \times (0.5) + (14,000 - 18,000)^2 \times (0.15) + (10,000 - 18,000)^2 \times (0.1)$

= 64,00,000 + 24,00,000 + 0 + 24,00,000 + 64,00,000 = 1,76,00,000

Standard Deviation ($\sigma$) = $\sqrt{\text{Variance}(\sigma^2)} = \sqrt{1,76,00,000} = 4195.23$

| Projects | Coefficient of variation | Risk | Expected Value |
|---|---|---|---|
| A | $\frac{2190.90}{14000} = 0.1565$ | Less | Less |
| B | $\frac{4195.23}{18000} = 0.2331$ | More | More |

In project A, risk per rupee of cash flow is ₹ 0.16 while in project B, it is ₹ 0.23. Therefore, Project A is better than Project B.

## SM Ch 3 · Illustration 7 (under 4.2.1 Risk Adjusted Discount Rate)

### Illustration 7

An enterprise is investing ₹ 100 lakhs in a project. The risk-free rate of return is 7%. Risk premium expected by the Management is 7%. The life of the project is 5 years. Following are the cash flows that are estimated over the life of the project:

| Year | Cash flows (₹ in lakhs) |
|---|---|
| 1 | 25 |
| 2 | 60 |
| 3 | 75 |
| 4 | 80 |
| 5 | 65 |

Calculate Net Present Value of the project based on Risk free rate and also on the basis of Risks adjusted discount rate.

## Solution — ICAI

**Solution**

The Present Value of the Cash Flows for all the years by discounting the cash flow at 7% is calculated as below:

| Year | Cash flows (₹ in lakhs) | Discounting Factor @ 7% | Present value of Cash Flows (₹ In Lakhs) |
|---|---|---|---|
| 1 | 25 | 0.935 | 23.38 |
| 2 | 60 | 0.873 | 52.38 |
| 3 | 75 | 0.816 | 61.20 |
| 4 | 80 | 0.763 | 61.04 |
| 5 | 65 | 0.713 | 46.35 |
| Total of Present value of Cash flows | | | 244.34 |
| Less: Initial investment | | | 100.00 |
| **Net Present Value (NPV)** | | | **144.34** |

Now, when the risk-free rate is 7% and the risk premium expected by the Management is 7%, then risk adjusted discount rate is 7% + 7% = 14%.

Discounting the above cash flows using the Risk Adjusted Discount Rate would be as below:

| Year | Cash flows (₹ in Lakhs) | Discounting Factor @ 14% | Present Value of Cash Flows (₹ in lakhs) |
|---|---|---|---|
| 1 | 25 | 0.877 | 21.93 |
| 2 | 60 | 0.769 | 46.14 |
| 3 | 75 | 0.675 | 50.63 |
| 4 | 80 | 0.592 | 47.36 |
| 5 | 65 | 0.519 | 33.74 |
| Total of Present value of Cash flows | | | 199.79 |
| Less: Initial investment | | | 100.00 |
| Net present value (NPV) | | | 99.79 |

**Advantages of Risk-adjusted discount rate**

(1) It is easy to understand.

(2) It incorporates risk premium in the discounting factor.

**Limitations of Risk-adjusted discount rate**

(1) Difficulty in finding risk premium and risk-adjusted discount rate.

(2) Though NPV can be calculated but it is not possible to calculate Standard Deviation of a given project.

## SM Ch 3 · Illustration 8 (under 4.2.2 Certainty Equivalent (CE))

**Illustration 8**

If Investment proposal costs ₹ 45,00,000 and risk free rate is 5%, calculate net present value under certainty equivalent technique.

| Year | Expected cash flow (₹) | Certainty Equivalent coefficient |
|---|---|---|
| 1 | 10,00,000 | 0.90 |
| 2 | 15,00,000 | 0.85 |
| 3 | 20,00,000 | 0.82 |
| 4 | 25,00,000 | 0.78 |

## Solution — ICAI

**Solution**

$$NPV = \frac{10,00,000\times(0.90)}{(1.05)} + \frac{15,00,000\times(0.85)}{(1.05)^2} + \frac{20,00,000\times(0.82)}{(1.05)^3} + \frac{25,00,000\times(0.78)}{(1.05)^4} - 45,00,000$$

= ₹ 5,34,570

**Advantages of Certainty Equivalent Method**

1. The certainty equivalent method is **simple and easy** to understand and apply.
2. It can **easily be calculated for different risk levels** applicable to different cash flows. For example, if in a particular year, a higher risk is associated with the cash flow, it can be easily adjusted and the NPV can be recalculated accordingly.

**Disadvantages of Certainty Equivalent Method**

1. There is **no objective** or mathematical method to estimate certainty equivalents. Certainty Equivalents are subjective and vary as per each individual's estimate.
2. Certainty equivalents are decided by the management based on their perception of risk. However, the **risk perception of the shareholders** who are the money lenders for the project is **ignored**. Hence, it is not used often in corporate decision making.

**Risk-adjusted Discount Rate Vs. Certainty-Equivalent**

Certainty Equivalent Method is superior to Risk Adjusted Discount Rate Method as it does not assume that risk increases with time at constant rate. Each year's Certainty Equivalent Coefficient is based on level of risk impacting its cash flow. Despite its soundness, it is not preferable like Risk Adjusted Discount Rate Method. It is difficult to specify a series of Certainty Equivalent Coefficients but simple to adjust discount rates.

## SM Ch 3 · Illustration 9 (under 4.3.1 Sensitivity Analysis)

**Illustration 9**

X Ltd. is considering its new project with the following details:

| Sr. No. | Particulars | Figures |
|---|---|---|
| 1 | Initial capital cost | ₹ 400 Cr. |
| 2 | Annual unit sales | 5 Cr. |
| 3 | Selling price per unit | ₹ 100 |
| 4 | Variable cost per unit | ₹ 50 |
| 5 | Fixed costs per year | ₹ 50 Cr. |
| 6 | Discount Rate | 6% |

Required:

1. Calculate the NPV of the project.
2. Compute the impact on the project's NPV considering a 2.5 per cent adverse variance in each variable. Which variable is having maximum effect?

Consider Life of the project as 3 years.

## Solution — ICAI

**Solution**

**1. Calculation of Net Cash Inflow per year**

| | Particulars | Amount (₹) |
|---|---|---|
| A | Selling price per unit | 100 |
| B | Variable cost per unit | 50 |
| C | Contribution per unit (A - B) | 50 |
| D | Number of units sold per year | 5 Cr. |
| E | Total Contribution (C × D) | ₹ 250 Cr. |
| F | Fixed cost per year | ₹ 50 Cr. |
| G | Net cash inflow per year (E - F) | ₹ 200 Cr. |

**Calculation of Net Present Value (NPV) of the Project**

| Year | Year Cash Flow (₹ in Cr.) | PV factor @ 6% | Present Value (PV) (₹ in Cr.) |
|---|---|---|---|
| 0 | (400.00) | 1.000 | (400.00) |
| 1 | 200.00 | 0.943 | 188.60 |
| 2 | 200.00 | 0.890 | 178.00 |
| 3 | 200.00 | 0.840 | 168.00 |
| Net Present Value | | | 134.60 |

Here, NPV represent the most likely outcomes and not the actual outcomes. The actual outcome can be lower or higher than the expected outcome.

**2. Sensitivity Analysis considering 2.5 % Adverse Variance in each variable**

| | Particulars | Base | Initial capital cost increased to ₹ 410 crore | Selling Price per Unit Reduced to ₹ 97.5 | Variable Cost Per Unit increased to ₹ 51.25 | Fixed Cost Per Unit increased to ₹ 51.25 | Units sold per year reduced to 4.875 crore |
|---|---|---|---|---|---|---|---|
| | | (₹) | (₹) | (₹) | (₹) | (₹) | (₹) |
| A | Selling price per unit | 100 | 100 | 97.5 | 100 | 100 | 100 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| B | Variable cost per unit | 50 | 50 | 50 | 51.25 | 50 | 50 |
| C | Contribution per unit (A - B) | 50 | 50 | 47.5 | 48.75 | 50 | 50 |
| | | (₹ in Cr.) | (₹ in Cr.) | (₹ in Cr.) | (₹ in Cr.) | (₹ in Cr.) | (₹ in Cr.) |
| D | Number of units sold per year (units in Crores) | 5 | 5 | 5 | 5 | 5 | 4.875 |
| E | Total Contribution (C × D) | 250 | 250 | 237.5 | 243.75 | 250 | 243.75 |
| F | Fixed cost per year | 50 | 50 | 50 | 50 | 51.25 | 50 |
| G | Net Cash Inflow per year (E - F) | 200 | 200 | 187.5 | 193.75 | 198.75 | 193.75 |
| H | PV of Net cash Inflow per year (G × 2.673) | 534.60 | 534.60 | 501.19 | 517.89 | 531.26 | 517.89 |
| I | Initial capital cost | 400 | 410 | 400 | 400 | 400 | 400 |
| J | NPV (H - I) | 134.60 | 124.60 | 101.19 | 117.89 | 131.26 | 117.89 |
| K | Percentage Change in NPV | - | -7.43% | -24.82% | -12.41% | -2.48% | -12.41% |

The above table shows that by changing one variable at a time by 2.5% (adverse) while keeping the others constant, the impact in percentage terms on the NPV of the project can be calculated. Thus, the change in selling price has the maximum effect on the NPV by 24.82%.

**Advantages of Sensitivity Analysis**:

Following are the main advantages of Sensitivity Analysis:

(1) **Critical Issues**: This analysis identifies critical factors that impinge on a project's success or failure.

(2) **Simplicity**: It is a simple technique.

**Disadvantage of Sensitivity Analysis**

Following are the main disadvantages of Sensitivity Analysis:

(1) **Assumption of Independence**: This analysis assumes that all variables are independent i.e. they are not related to each other, which is unlikely in real life.

(2) **Ignore probability**: This analysis does not look to the probability of changes in the variables.

## SM Ch 3 · Illustration 10 (under 4.3.2 Scenario Analysis)

### Illustration 10

XYZ Ltd. is considering a project "A" with an initial outlay of ₹ 14,00,000 and the possible three cash inflow attached with the project as follows:

| Particulars | Year 1 | Year 2 | Year 3 |
|---|---|---|---|
| Worst case | 450 | 400 | 700 |
| Most likely | 550 | 450 | 800 |
| Best case | 650 | 500 | 900 |

Assuming the cost of capital as 9%, determine NPV in each scenario. If XYZ Ltd is certain about the most likely result in first two years but uncertain about the third year's cash flow, analyze what will be the NPV expecting worst scenario in the third year.

## Solution — ICAI

**Solution**

The possible outcomes will be as follows:

| Year | PVF @ 9% | Worst Case | | Most likely | | Best case | |
|---|---|---|---|---|---|---|---|
| | | Cash Flow | PV | Cash Flow | PV | Cash Flow | PV |
| | | (₹ '000) | (₹ '000) | (₹ '000) | (₹ '000) | (₹ '000) | (₹ '000) |
| 0 | 1 | (1,400) | (1,400) | (1,400) | (1,400) | (1,400) | (1,400) |
| 1 | 0.917 | 450 | 412.65 | 550 | 504.35 | 650 | 596.05 |
| 2 | 0.842 | 400 | 336.80 | 450 | 378.90 | 500 | 421.00 |
| 3 | 0.772 | 700 | 540.40 | 800 | 617.60 | 900 | 694.80 |
| NPV | | | -110.15 | | 100.85 | | 311.85 |

If XYZ Ltd. is certain about the most likely result in first two years but uncertain about the third year's cash flow, then, NPV expecting worst case scenario is expected in the third year will be as follows:

$$= -₹\ 14,00,000 + \frac{₹\ 5,50,000}{(1+0.09)} + \frac{₹\ 4,50,000}{(1+0.09)^2} + \frac{₹\ 7,00,000}{(1+0.09)^3}$$

= − ₹ 14,00,000 + ₹ 5,04,587 + ₹ 3,78,756 + ₹ 5,40,528 = ₹ 23,871

**Scenario Analysis Vs Sensitivity Analysis**

- Sensitivity analysis and Scenario analysis both help to understand the impact of the change in input variable on the outcome of the project. However, there are certain basic differences between the two.
- Sensitivity analysis calculates the impact of the change of a single input variable on the outcome of the project viz., NPV or IRR. The sensitivity analysis thus enables to identify that single critical variable which can impact the outcome in a huge way and the range of outcomes of the project given the change in the input variable.
- Scenario analysis, on the other hand, is based on a scenario. The scenario may be recession or a boom wherein depending on the scenario, all input variables change. Scenario Analysis calculates the outcome of the project considering this scenario where the variables have changed simultaneously. Similarly, the outcome of the project would also be considered for the normal and recessionary situation. The variability in the outcome under the three different scenarios would help the management to assess the risk a project carries. Higher deviation in the outcome can be assessed as higher risk and lower to medium deviation can be assessed accordingly.

- Scenario analysis is far more complex than sensitivity analysis because in scenario analysis all inputs are changed simultaneously, considering the situation in hand while in sensitivity analysis, only one input is changed and others are kept constant.

## SM Ch 3 · Illustration 11 (under 4.3.4. Decision Tree Analysis)

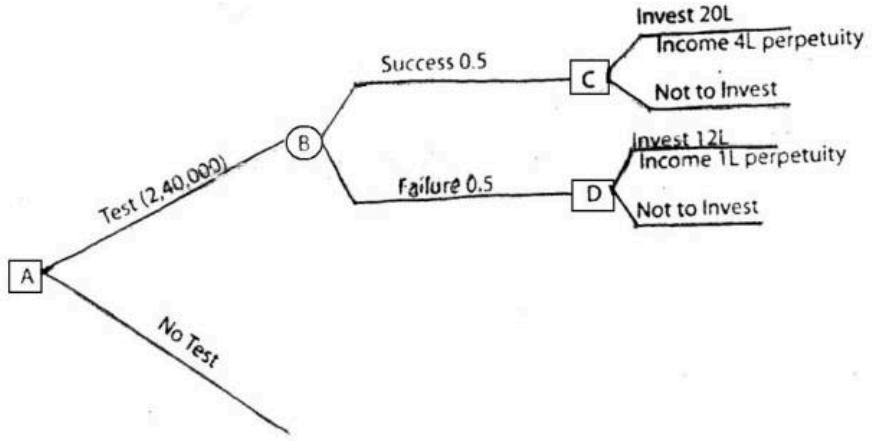
### Illustration 11

L & R Limited wishes to develop new virus-cleaner software. The cost of the pilot project would be ₹ 2,40,000. Presently, the chances of the product being successfully launched on a commercial scale are rated at 50%. In case it does succeed. L&R can invest a sum of ₹ 20 lacs to market the product. Such an effort can generate perpetually, an annual net after tax cash income of ₹ 4 lacs. Even if the commercial launch fails, they can make an investment of a smaller amount of ₹ 12 lacs with the hope of gaining perpetually a sum of ₹ 1 lac. Evaluate the proposal, adopting decision tree approach. The discount rate is 10%.

## Solution — ICAI

### Solution

Decision tree diagram is given below:




**Evaluation**

At Decision Point C: The choice is between investing ₹ 20 lacs for a perpetual benefit of ₹ 4 lacs and not to invest. The preferred choice is to invest, since the capitalized value of benefit of ₹ 4 lacs (at 10%) adjusted for the investment of ₹ 20 lacs, yields a net benefit of ₹ 20 lacs.

At Decision Point D: The choice is between investing ₹ 12 lacs, for a similar perpetual benefit of ₹ 1 lac. and not to invest. Here the invested amount is greater than capitalized value of benefit at ₹ 10 lacs. There is a negative benefit of ₹ 2 lacs. Therefore, it would not be prudent to invest.

At Outcome Point B: Evaluation of EMV is as under (₹ in lacs).

| Outcome | Amount (₹) | Probability | Result (₹) |
|---|---|---|---|
| Success | 20.00 | 0.50 | 10.00 |
| Failure | 0.00 | 0.50 | 00.00 |
| Net result | | | 10.00 |

EMV at B is, therefore, ₹10 lacs.

At A: Decision is to be taken based on preferences between two alternatives. The first is to test, by investing ₹2,40,000 and reap a benefit of ₹ 10 lacs. The second is not to test, and thereby losing the opportunity of a possible gain.

The preferred choice is, therefore, investing a sum of ₹ 2,40,000 and undertaking the test.

## 5. Replacement Decision

### SM Ch 3 · Illustration 12 (under 5.1 Replacement of Existing Machine)

**Illustration 12**

A Company named Roby's cube decided to replace the existing Computer system of their organisation. Original cost of old system was ₹ 25,000 and it was installed 5 years ago. Current market value of old system is ₹ 5,000. Depreciation of the old system was charged with life of 10 years with Estimated Salvage value as Nil. Depreciation of the new system will be charged with life over 5 years. Present cost of the new system is ₹ 50,000. Estimated Salvage value of the new system is ₹ 1,000. Estimated cost savings with new system is ₹ 5,000 per year. Increase in sales with new system is assumed at 10% per year based on original total sales of ₹ 10,00,000. Company follows straight line method of depreciation. Cost of capital of the company is 10% whereas tax rate is 30%.

### Solution — ICAI

**Solution**

**Step I.** Net cash outflow (assumed at current time) [Present values of cost]:

a. (Book value of old system – market value of old system) × Tax Rate

= Tax payable/savings from sale

= [(₹ 25,000 – 5 × ₹ 2,500) – ₹ 5,000] × 0.30 = ₹ 7,500 × 0.30

= ₹ 2,250

b. Cost of new system – [Tax payable/savings from sale + Market value of old system] = Net cash outflow

Or, ₹ 50,000 – [₹ 2,250 + ₹ 5,000] = ₹42,750

**Step II.** Estimated change in cash flows per year if replacement decision is implemented.

Change in cash flow = [(Change in sales ± Change in operating costs)-Change in depreciation)] (1-tax rate) + Change in depreciation

= [₹ 1,00,000 × 0.1 + ₹ 5,000 – (₹ 49,000/5 – ₹ 25,000/10)] (1-0.30) + (₹ 49,000/5 – ₹ 25000/10)]

= ₹ 12,690

**Step III.** Present value of benefits = Present value of yearly cash flows + Present value of estimated salvage of new system

= ₹ 12,690 × PVIFA (10%, 5) + ₹ 1,000 × PVIF (10%, 5)

= ₹ 48,723

**Step IV.** Net present value = Present value of benefits - Present value of costs

= ₹ 48,723 – ₹ 42,750

= ₹ 5,973

**Step V.** Decision rule: Since NPV is positive we should accept the proposal to replace the machine.

## SM Ch 3 · Illustration 13 (under 5.2 Optimum Replacement Cycle)

### Illustration 13

X Ltd. is a taxi operator. Each taxi cost to company ₹ 4,00,000 and has a useful life of 3 years. The taxi's operating cost for each of 3 years and salvage value at the end of year is as follows:

| | Year 1 | Year 2 | Year 3 |
|---|---|---|---|
| Operating Cost | ₹ 1,80,000 | ₹ 2,10,000 | ₹ 2,38,000 |
| Resale Value | ₹ 2,80,000 | ₹ 2,30,000 | ₹ 1,68,000 |

You are required to determine the optimal replacement period of taxi if cost of capital of X Ltd. is 10%.

## Solution — ICAI

### Solution

**NPV if taxi is kept for 1 Year**

= – ₹ 4,00,000 - ₹ 1,80,000 (0.909) + ₹ 2,80,000 (0.909)

= – ₹ 3,09,100

**NPV if taxi is kept for 2 Year**

= – ₹ 4,00,000 – ₹ 1,80,000 x 0.909 + ₹ 20,000 x 0.826

= – ₹ 5,47,100

**NPV if taxi is kept for 3 Year**

= – ₹ 4,00,000 – ₹ 1,80,000 x 0.909 – ₹ 2,10,000 x 0.826 – ₹ 70,000 x 0.751

= – ₹ 7,89,650

Since above NPV figures relate to different periods, there are not comparable. to make them comparable we shall use concept of EAC as follows:

EAC of 1 year

$$\frac{3,09,100}{0.909} = ₹\ 3,40,044$$

EAC of 2 year

$$\frac{5,47,100}{1.735} = ₹\ 3,15,331$$

EAC of 3 year

$$\frac{7,89,650}{2.486} = ₹\ 3,17,639$$

Since lowest EAC incur if taxi for 2 year; Hence the optimum replacement cycle to replace taxi in 2 years.

# Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

## Practical QI

1. Skylark Airways is planning to acquire a light commercial aircraft for flying class clients at an investment of ₹ 50,00,000. The expected cash flow after tax for the next three years is as follows: (₹)

| Year 1 | | Year 2 | | Year 3 | |
|---|---|---|---|---|---|
| CFAT | Probability | CFAT | Probability | CFAT | Probability |
| 14,00,000 | 0.1 | 15,00,000 | 0.1 | 18,00,000 | 0.2 |
| 18,00,000 | 0.2 | 20,00,000 | 0.3 | 25,00,000 | 0.5 |
| 25,00,000 | 0.4 | 32,00,000 | 0.4 | 35,00,000 | 0.2 |
| 40,00,000 | 0.3 | 45,00,000 | 0.2 | 48,00,000 | 0.1 |

The Company wishes to take into consideration all possible risk factors relating to airline operations. The company wants to know:

(i) The expected NPV of this venture assuming independent probability distribution with 6 per cent risk free rate of interest.

(ii) The possible deviation in the expected value.

(iii) How would standard deviation of the present value distribution help in Capital Budgeting decisions?

## Solution — ICAI

**1. (i) Expected NPV**

(₹ in lakhs)

| Year I | | | Year II | | | Year III | | |
|---|---|---|---|---|---|---|---|---|
| CFAT | P | CF×P | CFAT | P | CF×P | CFAT | P | CF×P |
| 14 | 0.1 | 1.4 | 15 | 0.1 | 1.5 | 18 | 0.2 | 3.6 |
| 18 | 0.2 | 3.6 | 20 | 0.3 | 6.0 | 25 | 0.5 | 12.5 |
| 25 | 0.4 | 10.0 | 32 | 0.4 | 12.8 | 35 | 0.2 | 7.0 |
| 40 | 0.3 | 12.0 | 45 | 0.2 | 9 | 48 | 0.1 | 4.8 |
| | $\bar{x}$ or $\overline{CF}$ | 27.0 | | $\bar{x}$ or $\overline{CF}$ | 29.3 | | $\bar{x}$ or $\overline{CF}$ | 27.9 |

| NPV | PV factor @ 6% | Total PV |
|---|---|---|
| 27 | 0.943 | 25.461 |
| 29.3 | 0.890 | 26.077 |
| 27.9 | 0.840 | 23.436 |
| | PV of cash inflow | 74.974 |
| | *Less*: Cash outflow | 50.000 |
| | NPV | 24.974 |

**(ii) Possible deviation in the expected value**

| *Year I* | | | | |
|---|---|---|---|---|
| $X - \bar{X}$ | $X - \bar{X}$ | $(X - \bar{X})^2$ | $P_1$ | $(X - \bar{X})^2 P^1$ |
| 14 – 27 | -13 | 169 | 0.1 | 16.9 |
| 18 – 27 | -9 | 81 | 0.2 | 16.2 |

| | | | | |
|---|---|---|---|---|
| 25 – 27 | -2 | 4 | 0.4 | 1.6 |
| 40 – 27 | 13 | 169 | 0.3 | 50.7 |
| | | | | 85.4 |

$\sigma_1 = \sqrt{85.4} = 9.241$

| Year II | | | | |
|---|---|---|---|---|
| $X - \bar{X}$ | $X - \bar{X}$ | $(X - \bar{X})^2$ | $P_2$ | $(X - \bar{X})^2 \times P_2$ |
| 15-29.3 | -14.3 | 204.49 | 0.1 | 20.449 |
| 20-29.3 | -9.3 | 86.49 | 0.3 | 25.947 |
| 32-29.3 | 2.7 | 7.29 | 0.4 | 2.916 |
| 45-29.3 | 15.7 | 246.49 | 0.2 | 49.298 |
| | | | | 98.61 |

$\sigma_2 = \sqrt{98.61} = 9.930$

| Year III | | | | |
|---|---|---|---|---|
| $X - \bar{X}$ | $X - \bar{X}$ | $(X - \bar{X})^2$ | $P_3$ | $(X - \bar{X})^2 \times P_3$ |
| 18-27.9 | -9.9 | 98.01 | 0.2 | 19.602 |
| 25-27.9 | -2.9 | 8.41 | 0.5 | 4.205 |
| 35-27.9 | 7.1 | 50.41 | 0.2 | 10.082 |
| 48-27.9 | 20.1 | 404.01 | 0.1 | 40.401 |
| | | | | 74.29 |

$\sigma_3 = \sqrt{74.29} = 8.619$

Standard deviation about the expected value:

$$\sigma = \sqrt{\frac{85.4}{(1.06)^2} + \frac{98.61}{(1.06)^4} + \frac{74.29}{(1.06)^6}} = 14.3696$$

**(iii)** Standard deviation is a statistical measure of dispersion; it measures the deviation from a central number i.e. the mean.

In the context of capital budgeting decisions especially where we take up two or more projects giving somewhat similar mean cash flows, by calculating standard deviation in such cases, we can measure in each case the extent of variation. It can then be used to identify which of the projects is least risky in terms of variability of cash flows.

A project, which has a lower coefficient of variation will be preferred if sizes are heterogeneous.

Besides this, if we assume that probability distribution is approximately normal we are able to calculate the probability of a capital budgeting project generating a net present value less than or more than a specified amount.

## Practical Q2

2. Cyber Company is considering two mutually exclusive projects. Investment outlay of both the projects is ₹ 5,00,000 and each is expected to have a life of 5 years. Under three possible situations their annual cash flows and probabilities are as under:

| | | Cash Flow (₹) | |
|---|---|---|---|
| Situation | Probabilities | Project A | ProjectB |
| Good | 0.3 | 6,00,000 | 5,00,000 |
| Normal | 0.4 | 4,00,000 | 4,00,000 |
| Worse | 0.3 | 2,00,000 | 3,00,000 |

The cost of capital is 7 per cent, which project should be accepted? Explain with workings.

## Solution — ICAI

**2. Project A**

Expected Net Cash flow (ENCF)

0.3 (6,00,000) + 0.4 (4,00,000) + 0.3 (2,00,000) = 4,00,000

$\sigma^2$=0.3 (6,00,000– 4,00,000)$^2$ + 0.4 (4,00,000 – 4,00,000)$^2$ + 0.3 (2,00,000 – 4,00,000)$^2$

$\sigma = \sqrt{24,00,00,00,000}$

$\sigma$ = 1,54,919.33

Present Value of Expected Cash Inflows = 4,00,000 × 4.100 = 16,40,000

NPV = 16,40,000 – 5,00,000 = 11,40,000

**Project B**

ENCF = 0.3 (5,00,000) + 0.4 (4,00,000) + 0.3 (3,00,000) = 4,00,000

$\sigma^2$=0.3 (5,00,000 – 4,00,000)$^2$ + 0.4 (4,00,000 – 4,00,000)$^2$ + 0.3 (3,00,000 – 4,00,000)$^2$

$\sigma = \sqrt{6,00,00,00,000}$

$\sigma$ = 77,459.66

Present Value of Expected Cash Inflows = 4,00,000 × 4.100 = 16,40,000

NPV = 16,40,000 – 5,00,000 = 11,40,000

**Recommendation:** NPV in both projects being the same, the project should be decided on the basis of standard deviation and hence project 'B' should be accepted having lower standard deviation, means less risky.

## Practical Q3

3. A company is considering Projects X and Y with following information:

| Project | Expected NPV (₹) | Standard deviation |
|---|---|---|
| X | 1,22,000 | 90,000 |
| Y | 2,25,000 | 1,20,000 |

(i) Which project will you recommend based on the above data?

(ii) Explain whether your opinion will change, if you use coefficient of variation as a measure of risk.

(iii) Which measure is more appropriate in this situation and why?

## Solution — ICAI

**3. (i)** On the basis of standard deviation project X be chosen because it is less risky than Project Y having higher standard deviation.

**(ii)** $CV_x = \frac{SD}{ENPV} = \frac{90,000}{1,22,000} = 0.738$

$CV_y = \frac{1,20,000}{2,25,000} = 0.533$

On the basis of Co-efficient of Variation (C.V.) Project X appears to be riskier and hence Y should be accepted.

**(iii)** However, the NPV method in such conflicting situation is best because the NPV method is in compatibility of the objective of wealth maximisation in terms of time value.

## Practical Q4

4. KLM Ltd., is considering taking up one of the two projects-Project-K and Project-So Both the projects having same life require equal investment of ₹ 80 lakhs each. Both are estimated to have almost the same yield. As the company is new to this type of business, the cash flow arising from the projects cannot be estimated with certainty. An attempt was therefore, made to use probability to analyse the pattern of cash flow from other projects during the first year of operations. This pattern is likely to continue during the life of these projects. The results of the analysis are as follows:

| Project K | | Project S | |
|---|---|---|---|
| Cash Flow (in ₹) | Probability | Cash Flow (in ₹) | Probability |
| 11 | 0.10 | 09 | 0.10 |
| 13 | 0.20 | 13 | 0.25 |
| 15 | 0.40 | 17 | 0.30 |
| 17 | 0.20 | 21 | 0.25 |
| 19 | 0.10 | 25 | 0.10 |

**Required:**

(i) Calculate variance, standard deviation and co-efficient of variance for both the projects.

(ii) Which of the two projects is riskier?

## Solution — ICAI

**4. Calculation of Variance and Standard Deviation**

**Project K**

Expected Net Cash Flow

= (0.10 x 11) + (0.20 x13) + (0.40 x 15) + (0.20 x 17) + (0.10 x 19)

= 1.1 + 2.6 + 6 + 3.4 + 1.9 = 15

$$\sigma^2 = 0.10(11-15)^2 + 0.20(13-15)^2 + 0.40(15-15)^2 + 0.20(17-15)^2 + 0.10(19-15)^2$$

= 1.6 + 0.8 + 0 + 0.8 + 1.6 = 4.8

$$\sigma = \sqrt{4.8} = 2.19$$

**Project S**

Expected Net Cash Flow

= (0.10 X 9) + (0.25 X 13) + (0.30 X 17) + (0.25 X 21) + (0.10 X 25)

= 0.9 + 3.25 + 5.1 + 5.25 + 2.5 = 17

$$\sigma^2 = 0.1(9-17)^2 + 0.25(13-17)^2 + 0.30(17-17)^2 + 0.25(21-17)^2 + 0.10(25-17)^2$$

= 6.4 + 4 + 0 + 4 + 6.4 = 20.8

$$\sigma = \sqrt{20.8} = 4.56$$

**Calculation of Coefficient of Variation**

$$\text{Coefficient of Variation} = \frac{\text{Standard Deviation}}{\text{Mean}}$$

$$\text{Project K} = \frac{2.19}{15} = 0.146$$

$$\text{Project S} = \frac{4.56}{17} = 0.268$$

Project S is riskier as it has higher Coefficient of Variation.

## Practical QS

5. Project X and Project Y are under the evaluation of XY Co. The estimated cash flows and their probabilities are as below:

Project X : Investment (year 0) ₹ 70 lakhs

| Probability weights | 0.30 | 0.40 | 0.30 |
|---|---|---|---|
| Years | ₹ lakhs | ₹ lakhs | ₹ lakhs |
| 1 | 30 | 50 | 65 |
| 2 | 30 | 40 | 55 |
| 3 | 30 | 40 | 45 |

Project Y: Investment (year 0) ₹ 80 lakhs.

| Probability weighted | Annual cash flows through life |
|---|---|
| | ₹ lakhs |
| 0.20 | 40 |
| 0.50 | 45 |
| 0.30 | 50 |

(a) Which project is better based on NPV, criterion with a discount rate of 10%?

(b) Compute the standard deviation of the present value distribution and analyse the inherent risk of the projects.

## Solution — ICAI

**5. (a) Calculation of NPV of XY Co.:**

| Project X | | Cash flow | PVF | PV |
|---|---|---|---|---|
| Year | | | | |
| 1 | (30 × 0.3) + (50 × 0.4) + (65 × 0.3) | 48.5 | 0.909 | 44.09 |
| 2 | (30 × 0.3) + (40 × 0.4) + (55 × 0.3) | 41.5 | 0.826 | 34.28 |
| 3 | (30 × 0.3) + (40 × 0.4) + (45 × 0.3) | 38.5 | 0.751 | 28.91 |
| | | | | 107.28 |
| | NPV: (107.28 – 70.00) = | | | (+) 37.28 |

**Project Y (For 1-3 Years)**

| | | | | |
|---|---|---|---|---|
| 1-3 | (40 × 0.2) + (45 × 0.5) + (50 × 0.3) | 45.5 | 2.487 | 113.16 |
| | NPV (113.16 – 80.00) | | | (+) 33.16 |

**(b) Calculation of Standard deviation σ**

As per Hiller's model

$$M = \sum_{i=0}^{n} (1+r)^{-1} Mi$$

$$\sigma^2 = \sum_{i=0}^{n} (1+r)^{-2i} \sigma_i^2$$

Hence

Project X

Year

1 $\sqrt{(30-48.5)^2 0.30 + (50-48.5)^2 0.40 + (65-48.5)^2 0.30} = \sqrt{185.25} = 13.61$

2 $\sqrt{(30-41.5)^2 0.30 + (40-41.5)^2 0.40 + (55-41.5)^2 0.30} = \sqrt{95.25} = 9.76$

3 $\sqrt{(30-38.5)^2 0.30 + (40-38.5)^2 0.40 + (45-38.5)^2 0.30} = \sqrt{35.25} = 5.94$

Standard Deviation about the expected value

$$= \sqrt{\frac{185.25}{(1+0.10)^2} + \frac{95.25}{(1+0.10)^4} + \frac{35.25}{(1+0.10)^6}}$$

$$= \sqrt{\frac{185.25}{1.21} + \frac{95.25}{1.4641} + \frac{35.25}{1.7716}} = \sqrt{153.10+65.06+19.90}$$

$$= \sqrt{238.06} = 15.43$$

Project Y (For 1-3 Years)

$$\sqrt{(40-45.5)^2 0.20 + (45-45.5)^2 0.50 + (50-45.5)^2 0.30} \quad = \sqrt{12.25} \quad = 3.50$$

Standard Deviation about the expected value

$$= \sqrt{\frac{12.25}{(1+0.10)^2} + \frac{12.25}{(1+0.10)^4} + \frac{12.25}{(1+0.10)^6}}$$

$$= \sqrt{\frac{12.25}{1.21} + \frac{12.25}{1.4641} + \frac{12.25}{1.7716}} = \sqrt{10.12+8.37+6.91}$$

$$= \sqrt{25.4} = 5.03$$

***Analysis:*** Project Y is less risky as its Standard Deviation is less than Project X.

## Practical Q6

6. Shivam Ltd. is considering two mutually exclusive projects A and B. Project A costs ₹ 36,000 and project B ₹ 30,000. You have been given below the net present value probability distribution for each project.

| Project A | | Project B | |
|---|---|---|---|
| NPV estimates (₹) | Probability | NPV estimates (₹) | Probability |
| 15,000 | 0.2 | 15,000 | 0.1 |
| 12,000 | 0.3 | 12,000 | 0.4 |
| 6,000 | 0.3 | 6,000 | 0.4 |
| 3,000 | 0.2 | 3,000 | 0.1 |

(i) Compute the expected net present values of projects A and B.

(ii) Compute the risk attached to each project i.e. standard deviation of each probability distribution.

(iii) Compute the profitability index of each project.

(iv) Which project do you recommend? State with reasons.

## Solution — ICAI

**6. (i) Statement showing computation of expected net present value of Projects A and B:**

| Project A | | | Project B | | |
|---|---|---|---|---|---|
| NPV Estimate (₹) | Probability | Expected Value | NPV Estimate | Probability | Expected Value |
| 15,000 | 0.2 | 3,000 | 15,000 | 0.1 | 1,500 |
| 12,000 | 0.3 | 3,600 | 12,000 | 0.4 | 4,800 |
| 6,000 | 0.3 | 1,800 | 6,000 | 0.4 | 2,400 |
| 3,000 | 0.2 | 600 | 3,000 | 0.1 | 300 |
| | 1.0 | EV = 9,000 | | 1.0 | EV = 9,000 |

**(ii) Computation of Standard deviation of each project**

**Project A**

| P | X | (X – EV) | P (X - EV)² |
|---|---|---|---|
| 0.2 | 15,000 | 6,000 | 72,00,000 |
| 0.3 | 12,000 | 3,000 | 27,00,000 |
| 0.3 | 6,000 | - 3,000 | 27,00,000 |
| 0.2 | 3,000 | - 6,000 | 72,00,000 |
| | | | Variance = 1,98,00,000 |

Standard Deviation of Project A = $\sqrt{1,98,00,000}$ = ₹ 4,450

**Project B**

| P | X | (X – EV) | P (X - EV)² |
|---|---|---|---|
| 0.1 | 15,000 | 6,000 | 36,00,000 |
| 0.4 | 12,000 | 3,000 | 36,00,000 |
| 0.4 | 6,000 | - 3,000 | 36,00,000 |
| 0.1 | 3,000 | - 6,000 | 36,00,000 |
| | | | Variance = 1,44,00,000 |

Standard Deviation of Project A = $\sqrt{1,44,00,000}$ = ₹ 3,795

**(iii) Computation of profitability of each project**

Profitability index = Discount cash inflow / Initial outlay

In case of Project A : PI = $\frac{9,000+36,000}{36,000}=\frac{45,000}{36,000}=1.25$

In case of Project B : PI = $\frac{9,000+30,000}{30,000}=\frac{39,000}{30,000}=1.30$

**(iv)** Measurement of risk is made by the possible variation of outcomes around the expected value and the decision will be taken in view of the variation in the expected value where two projects have the same expected value, the decision will be the project which has smaller variation in expected value. In the selection of one of the two projects A and B, Project B is preferable because the possible profit which may occur is subject to less variation (or dispersion). Much higher risk is lying with project A.

## Practical Q7

7. Following are the estimates of the net cash flows and probability of a new project of M/s X Ltd.:

| | Year | P = 0.3 | P = 0.5 | P = 0.2 |
|---|---|---|---|---|
| Initial investment | 0 | 4,00,000 | 4,00,000 | 4,00,000 |
| Estimated net after tax cash inflows per year | 1 to 5 | 1,00,000 | 1,10,000 | 1,20,000 |
| Estimated salvage value (after tax) | 5 | 20,000 | 50,000 | 60,000 |

Required rate of return from the project is 10%. Find:

(i) The expected NPV of the project.

(ii) The best case and the worst case NPVs.

(iii) The probability of occurrence of the worst case if the cash flows are perfectly dependent overtime and independent overtime.

(iv) Standard deviation and coefficient of variation assuming that there are only three streams of cash flow, which are represented by each column of the table with the given probabilities.

(v) Coefficient of variation of X Ltd. on its average project which is in the range of 0.95 to 1.0. If the coefficient of variation of the project is found to be less risky than average, 100 basis points are deducted from the Company's cost of Capital

Should the project be accepted by X Ltd?

## Solution — ICAI

**7. (a) (i) Expected cash flows:-**

| Year | | | Net cash flows | P.V. | PV. @ 10% |
|---|---|---|---|---|---|
| 0 | (4,00,000 x 1) | = | (-)4,00,000 | 1.000 | (-)4,00,000 |
| 1 to 4 | (1,00,000x0.3+1,10,000x0.5 + 1,20,000 x 0.2) | = | 1,09,000 | 3.170 | 3,45,530 |
| 5 | [1,09,000 + (20,000 x 0.3 + 50,000 x 0.5 + 60,000 x 0.2)] | = | 1,52,000 | 0.621 | 94,392 |
| | NPV = | | | | 39,922 |

**(ii) ENPV of the worst case**

1,00,000 x 3.790 = ₹ 3,79,000 (Students may have 3.791 also the values will change accordingly)

20,000 x 0.621 = ₹ 12,420/-

ENPV = (-) 4,00,000 + 3,79,000 + 12,420 = (-) ₹ 8,580/-

ENPV of the best case

ENPV = (-) 4,00,000 + 1,20,000 x 3.790 + 60,000 x 0.621 = ₹ 92,060/-.

**(iii)** (a) Required probability = 0.3

(b) Required probability = $(0.3)^5 = 0.00243$

**(iv)** The base case NPV = (-) 4,00,000 + (1,10,000 x 3.79) + (50,000 x 0.621)

= ₹ 47,950/-

ENPV = 0.30 x (-) 8580 + 0.5 x 47950 + 92060 x 0.20 = ₹ 39,813/-

Therefore,

$$\sigma_{ENPV} = \sqrt{0.3(-8580-39{,}813)^2 + 0.5(47950-39813)^2 + 0.2(92{,}060-39{,}813)^2}$$

= ₹ 35,800/-

Therefore, CV = 35,800/39,813 = 0.90

**(v)** Risk adjusted out of cost of capital of X Ltd. = 10% - 1% = 9%.

NPV

| Year | Expected net cash flow | PV @ 9% | |
|---|---|---|---|
| 0 | (-) 4,00,000 | 1.000 | (-) 4,00,000 |
| 1 to 4 | 1,09,000 | 3.240 | 3,53,160 |
| 5 | 1,52,000 | 0.650 | 98,800 |
| | | ENPV = | 51,960 |

Therefore, the project should be accepted.

## Practical Q8

8. XY Ltd. has under its consideration a project with an initial investment of ₹ 1,00,000. Three probable cash inflow scenarios with their probabilities of occurrence have been estimated as below:

| Annual cash inflow (₹) | 20,000 | 30,000 | 40,000 |
|---|---|---|---|
| Probability | 0.1 | 0.7 | 0.2 |

The project life is 5 years and the desired rate of return is 20%. The estimated terminal values for the project assets under the three probability alternatives, respectively, are ₹ 0, 20,000 and 30,000.

You are required to:

(i) Find the probable NPV;

(ii) Find the worst-case NPV and the best-case NPV; and

(iii) State the probability occurrence of the worst case, if the cash flows are perfectly positively correlated over time.

## Solution — ICAI

**8.** The expected cash flows of the project are as follows:

| Year | Pr = 0.1 | Pr = 0.7 | Pr = 0.2 | Total |
|---|---|---|---|---|
| | ₹ | ₹ | ₹ | ₹ |
| 0 | -10,000 | -70,000 | -20,000 | -1,00,000 |
| 1 | 2,000 | 21,000 | 8,000 | 31,000 |
| 2 | 2,000 | 21,000 | 8,000 | 31,000 |
| 3 | 2,000 | 21,000 | 8,000 | 31,000 |
| 4 | 2,000 | 21,000 | 8,000 | 31,000 |
| 5 | 2,000 | 21,000 | 8,000 | 31,000 |
| 5 | 0 | 14,000 | 6,000 | 20,000 |

(i) NPV based on expected cash flows would be as follows:

$$=-₹\ 1{,}00{,}000 + \frac{₹\ 31{,}000}{(1+0.20)^1} + \frac{₹\ 31{,}000}{(1+0.20)^2} + \frac{₹\ 31{,}000}{(1+0.20)^3} + \frac{₹\ 31{,}000}{(1+0.20)^4} + \frac{₹\ 31{,}000}{(1+0.20)^5} + \frac{₹\ 20{,}000}{(1+0.20)^5}$$

= - ₹ 1,00,000 + ₹ 25,833.33 + ₹ 21,527.78 + ₹ 17,939.81 + ₹ 14,949.85+ ₹ 12,458.20 + ₹ 8,037.55

NPV = ₹ 746.52

(ii) For the worst case, the cash flows from the cash flow column farthest on the left are used to calculate NPV

$$= - ₹\ 100{,}000 + \frac{₹ 20{,}000}{(1+0.20)^1} + \frac{₹ 20{,}000}{(1+0.20)^2} + \frac{₹ 20{,}000}{(1+0.20)^3} + \frac{₹ 20{,}000}{(1+0.20)^4} + \frac{₹ 20{,}000}{(1+0.20)^5}$$

= - ₹ 100,000 + ₹ 16,666.67+ ₹ 13,888.89 + ₹ 11,574.07 + ₹ 9,645.06+ ₹ 8037.76

NPV = - ₹ 40,187.76

For the best case, the cash flows from the cash flow column farthest on the right are used to calculated NPV

$$= -₹\ 100{,}000 + \frac{₹\ 40{,}000}{(1+0.20)^1} + \frac{₹\ 40{,}000}{(1+0.20)^2} + \frac{₹\ 40{,}000}{(1+0.20)^3} + \frac{₹\ 40{,}000}{(1+0.20)^4} + \frac{₹\ 40{,}000}{(1+0.20)^5} + \frac{₹\ 30{,}000}{(1+0.20)^5}$$

= - ₹ 1,00,000 + ₹ 33,333.33+ ₹ 27,777.78 + ₹ 23,148.15+ ₹ 19,290.12 + ₹ 16,075.10 + ₹ 12,056.33

NPV = ₹ 31,680.81

(iii) If the cash flows are perfectly dependent, then the low cash flow in the first year will mean a low cash flow in every year. Thus, the possibility of the worst case occurring is the probability of getting ₹ 20,000 net cash flow in year 1 is 10%.

## Practical Q9

9. XYZ Ltd. is considering a project for which the following estimates are available:

| | ₹ |
|---|---|
| Initial Cost of the project | 10,00,000 |
| Sales price/unit | 60 |
| Cost/unit | 40 |
| Sales volumes | |
| Year 1 | 20000 units |
| Year 2 | 30000 units |
| Year 3 | 30000 units |

Discount rate is 10% p.a.

You are required to measure the sensitivity of the project in relation to each of the following parameters:

(a) Sales Price/unit

(b) Unit cost

(c) Sales volume

(d) Initial outlay and

(e) Project lifetime

Taxation may be ignored.

## Solution — ICAI

**9. Calculation of NPV**

$$NPV = -10,00,000 + \frac{20,000\times20}{1.1} + \frac{30,000\times20}{1.21} + \frac{30,000\times20}{1.331}$$

= - ₹ 10,00,000 + ₹ 3,63,636 + ₹ 4,95,868 + ₹ 4,50,789

= ₹ 13,10,293 – ₹ 10,00,000

= ₹ 3,10,293

**Measurement of sensitivity is as follows:**

**(a) Sales Price:-**

Let the sale price/Unit be S so that the project would break even with 0 NPV.

$$\therefore 10,00,000 = \frac{20,000\times(S-40)}{1.1} + \frac{30,000\times(S-40)}{1.21} + \frac{30,000(S-40)}{1.331}$$

S – 40 = ₹ 10,00,000/₹ 65,514

S – 40 = ₹ 15.26

S = ₹ 55.26 which represents a fall of (60-55.26)/60

Or 0.079 or 7.9%

**Alternative Method**

$$\frac{10,00,000 \times 20}{13,10,293} = ₹\ 15.26$$

S= ₹ 40 + ₹ 15.26

= ₹ 55.26

**Alternative Solution**

If sale Price decreased by say 10%, then NPV (at Sale Price of ₹ 60 – ₹ 6 = ₹ 54)

$$NPV = -10,00,000 + \frac{20000 \times 14}{(1.1)^1} + \frac{30000 \times 14}{(1.1)^2} + \frac{30000 \times 14}{(1.1)^3}$$

= -₹ 10,00,000 + ₹ 2,54,545 + ₹ 3,47,107 + ₹ 3,15,552

= - ₹ 82,796

$$\text{NPV decrease (\%)} = \frac{3,10,293-(-82,796)}{3,10,293} \times 100 = 126.68\%$$

**(b) Unit Cost:-**

If sales price = ₹ 60 the cost price required to give a margin of ₹ 15.26 is (₹ 60 – ₹ 15.26) or ₹ 44.74 which would represent a rise of 11.85% i.e., $\left(\frac{44.74-40}{40} \times 100\right)$

**Alternative Solution**

If unit cost increased by say 10%. The new NPV will be as follows:

$$NPV = -10,00,000 + \frac{20000 \times 16}{(1.1)^1} + \frac{30000 \times 16}{(1.1)^2} + \frac{30000 \times 16}{(1.1)^3}$$

= -₹ 10,00,000 + ₹ 2,90,909 + ₹ 3,96,694 + ₹ 3,60,631

= ₹ 48,234

$$\text{NPV decrease (\%)} = \frac{3,10,293-(48,234)}{3,10,293} \times 100 = 84.46\%$$

**(c) Sales volume:-**

The requisite percentage fall is:-

3,10,293/13,10,293 × 100 = 23.68%

**Alternative Solution**

If sale volume decreased by say 10%. The new NPV will be as follows:

$$NPV = -10,00,000 + \frac{18000\times20}{(1.1)^1} + \frac{27000\times20}{(1.1)^2} + \frac{27000\times20}{(1.1)^3}$$

= - ₹ 10,00,000 + ₹ 3,27,272 + ₹ 4,46,281 + ₹ 4,05,710

= ₹ 1,79,263

$$\text{NPV decrease (\%)} = \frac{3,10,293 - 1,79,263}{3,10,293} \text{ X } 100 = 42.22\%$$

**(d) Since PV of inflows remains at ₹13,10,293 the initial outlay must also be the same.**

∴ Percentage rise = 3,10,293/10,00,000 × 100 = 31.03%.

**Alternative Solution**

If initial outlay increased by say 10%. The new NPV will be as follows:

$$NPV = -₹\ 11,00,000 + \frac{20000\times20}{(1.1)^1} + \frac{30000\times20}{(1.1)^2} + \frac{30000\times20}{(1.1)^3}$$

= - ₹ 11,00,000 + ₹ 3,63,636 + ₹ 4,95,868 + ₹ 4,50,789 = ₹ 2,10,293

$$\text{NPV decrease (\%)} = \frac{3,10,293 - 2,10,293}{3,10,293} \text{ X } 100 = 32.22\%$$

**(e) Present value for 1st two years.**

= - ₹ 10,00,000 + ₹ 4,00,000 x 0.909 + ₹ 6,00,000 x 0.826

= - ₹ 10,00,000 + ₹ 3,63,600 + ₹ 4,95,600

= - ₹ 10,00,000 + ₹ 8,59,200

= - ₹ 1,40,800

∴ The project needs to run for some part of the third year so that the present value of return is ₹ 1,40,800. It can be computed as follows:

(i) 30,000 units x ₹ 20 x 0.751 = ₹ 4,50,600

(ii) Per day Production in (₹) assuming a year of 360 days =

$$\frac{₹\ 4,50,600}{360} = ₹\ 1,252$$

(iii) Days needed to recover ₹1,40,800 = $\frac{₹\ 1,40,800}{₹\ 1,252} = 112$

Thus, if the project runs for 2 years and 112 days then break even would be achieved representing a fall of $\frac{(3 - 2.311)}{3} \times 100 = 22.97\%$.

## Practical Q10

10. From the following details relating to a project, analyse the sensitivity of the project to changes in initial project cost, annual cash inflow and cost of capital:

| | |
|---|---|
| Initial Project Cost (₹) | 1,20,000 |
| Annual Cash Inflow (₹) | 45,000 |
| Project Life (Years) | 4 |
| Cost of Capital | 10% |

To which of the three factors, the project is most sensitive? (Use annuity factors: for 10% 3.169 and 11% 3.103).

## Solution — ICAI

**10. CALCULATION OF NPV**

| | ₹ |
|---|---|
| PV of cash inflows (₹ 45,000 x 3.169) | 1,42,605 |
| Initial Project Cost | 1,20,000 |
| NPV | 22,605 |
| **If initial project cost is varied adversely by 10%*** | |
| NPV (Revised) (₹ 1,42,605 - ₹ 1,32,000) | ₹ 10,605 |
| Change in NPV (₹ 22,605 – ₹ 10,605)/ ₹ 22,605 i.e. | 53.08 % |
| **If annual cash inflow is varied adversely by 10%*** | |
| Revised annual inflow | ₹ 40,500 |
| NPV (Revised) (₹ 40,500 x 3.169) – (₹ 1,20,000) | (+) ₹ 8,345 |
| Change in NPV (₹ 22,605 – ₹ 8,345) / ₹ 22,605 | 63.08 % |
| **If cost of capital is varied adversely by 10%*** | |
| NPV (Revised) (₹ 45,000 x 3.103) – ₹ 1,20,000 | (+) ₹ 19,635 |
| Change in NPV (₹ 22,605 – ₹ 19,635) / ₹ 22,605 | 13.14 % |

**Conclusion:** Project is most sensitive to 'annual cash inflow'.

*Note: Students may please note that they may assume any other percentage rate other than 10 % say 15%, 20 % 25 % etc.

## Practical Q11

11. Red Ltd. is considering a project with the following Cash flows:

₹

| Years | Cost of Plant | Recurring Cost | Savings |
|---|---|---|---|
| 0 | 10,000 | | |
| 1 | | 4,000 | 12,000 |
| 2 | | 5,000 | 14,000 |

The cost of capital is 9%. Measure the sensitivity of the project to changes in the levels of plant value, running cost and savings (considering each factor at a time) such that the NPV becomes zero. The P.V. factor at 9% are as under:

| Year | Factor |
|---|---|
| 0 | 1 |
| 1 | 0.917 |
| 2 | 0.842 |

Which factor is the most sensitive to affect the acceptability of the project?

## Solution — ICAI

**11. P.V. of Cash Flows**

| | | | |
|---|---|---|---|
| Year 1 | Running Cost | ₹ 4,000 x 0.917 | = (₹ 3,668) |
| | Savings | ₹ 12,000 x 0.917 | = ₹ 11,004 |
| Year 2 | Running Cost | ₹ 5,000 x 0.842 | = (₹ 4,210) |
| | Savings | ₹ 14,000 x 0.842 | = ₹ 11,788 |
| | | | ₹ 14,914 |
| Year 0 | *Less:* P.V. of Cash Outflow | ₹ 10,000 x 1 | ₹ 10,000 |
| | | NPV | ₹ 4,914 |

**Sensitivity Analysis**

(i) Increase of Plant Value by ₹ 4,914

$$\therefore \frac{4,914}{10,000} \times 100 = 49.14\%$$

(ii) Increase in PV of Running Cost by ₹ 4,914

$$\frac{4,914}{3,668+4,210} = \frac{4,914}{7,878} \times 100 = 62.38\%$$

(iii) Fall in PV of Saving by ₹ 4,914

$$\frac{4,914}{11,004+11,788} = \frac{4,914}{22,792} \times 100 = 21.56\%$$

Hence, savings factor is the most sensitive to affect the acceptability of the project as in comparison of other two factors a slight % change in this fact shall more affect the NPV than others.

**Alternative Solution**

P.V. of Cash Flows

| | | | |
|---|---|---|---|
| Year 1 | Running Cost | ₹ 4,000 x 0.917 | = (₹ 3,668) |
| | Savings | ₹ 12,000 x 0.917 | = ₹ 11,004 |
| Year 2 | Running Cost | ₹ 5,000 x 0.842 | = (₹ 4,210) |
| | Savings | ₹ 14,000 x 0.842 | = ₹ 11,788 |
| | | | ₹ 14,914 |
| Year 0 | Less: P.V. of Cash Outflow | ₹ 10,000 x 1 | ₹ 10,000 |
| | | NPV | ₹ 4,914 |

**Sensitivity Analysis**

(i) If the initial project cost is varied adversely by say 10%*.

NPV (Revised) (₹ 4,914 – ₹ 1,000) = ₹ 3,914

$$\text{Change in NPV } \frac{₹\ 4,914 - ₹\ 3,914}{₹\ 4,914} = 20.35\%$$

(ii) If Annual Running Cost is varied by say 10%*.

NPV (Revised) (₹ 4,914 – ₹ 400 X 0.917 – ₹ 500 X 0.842)

= ₹ 4,914 – ₹ 367 – ₹ 421= ₹ 4,126

Change in NPV $\frac{₹\ 4,914 - ₹\ 4,126}{₹\ 4,914}$ = 16.04%

(iii) If Saving is varied by say 10%*.

NPV (Revised) (₹ 4,914 – ₹ 1,200 X 0.917 – ₹ 1,400 X 0.842)

= ₹ 4,914 – ₹ 1,100 – ₹ 1,179 = ₹ 2,635

Change in NPV $\frac{4,914 - 2,635}{4,914}$ = 46.38%

Hence, savings factor is the most sensitive to affect the acceptability of the project.

* Any percentage of variation other than 10% can also be assumed.

## Practical Q12

12. The Easygoing Company Limited is considering a new project with initial investment, for a product "Survival". It is estimated that IRR of the project is 16% having an estimated life of 5 years.

Financial Manager has studied that project with sensitivity analysis and informed that annual fixed cost sensitivity is 7.8416%, whereas cost of capital (discount rate) sensitivity is 60%.

Other information available are:

Profit Volume Ratio (P/V) is 70%,

Variable cost ₹ 60/- per unit

Annual Cash Flow ₹ 57,500/-

Ignore Depreciation on initial investment and impact of taxation.

Calculate

(i) Initial Investment of the Project

(ii) Net Present Value of the Project

(iii) Annual Fixed Cost

(iv) Estimated annual unit of sales

(v) Break Even Units

Cumulative Discounting Factor for 5 years

| 8% | 9% | 10% | 11% | 12% | 13% | 14% | 15% | 16% | 17% | 18% |
|---|---|---|---|---|---|---|---|---|---|---|
| 3.993 | 3.890 | 3.791 | 3.696 | 3.605 | 3.517 | 3.433 | 3.352 | 3.274 | 3.199 | 3.127 |

## Solution — ICAI

**12. (i) Initial Investment**

IRR = 16% (Given)

At IRR, NPV shall be zero, therefore

Initial Cost of Investment = PVAF (16%,5) x Cash Flow (Annual)

= 3.274 x ₹ 57,500

= ₹ 1,88,255

**(ii) Net Present Value (NPV)**

Let Cost of Capital be X, then $\frac{16-X}{X} = 60\%$ X = 10%

Thus NPV of the project

= Annual Cash Flow x PVAF (10%, 5) – Initial Investment

= ₹ 57,500 x 3.791 – ₹ 1,88,255

= ₹ 2,17,982.50 – ₹ 1,88,255 = ₹ 29,727.50

**(iii) Annual Fixed Cost**

Let change in the Fixed Cost which makes NPV zero is X. Then,

₹ 29,727.50 – 3.791X = 0

Thus X = ₹ 7,841.60

Let original Fixed Cost be Y then,

Y × 7.8416% = ₹ 7,841.60

Y = ₹ 1,00,000

Thus Fixed Cost is equal to ₹ 1,00,000

**(iv) Estimated Annual Units of Sales**

$$\text{Selling Price per unit} = \frac{₹\ 60}{100\% - 70\%} = ₹\ 200$$

$$\frac{\text{Annual Cash Flow + Fixed Cost}}{\text{P/V Ratio}} = \text{Sales Value}$$

$$\frac{₹\ 57,500 + ₹\ 1,00,000}{0.70} = ₹\ 2,25,000$$

$$\text{Sales in Units} = \frac{₹\ 2,25,000}{₹\ 200} = 1,125 \text{ units}$$

**(v) Break Even Units**

$$\frac{\text{Fixed Cost}}{\text{Contribution Per Unit}} = \frac{1,00,000}{140} = 714.285 \text{ units}$$

## Practical Q13

13. Unnat Ltd. is considering investing ₹ 50,00,000 in a new machine. The expected life of machine is five years and has no scrap value. It is expected that 2,00,000 units will be produced and sold each year at a selling price of ₹ 30.00 per unit. It is expected that the variable costs to be ₹ 16.50 per unit and fixed costs to be ₹ 10,00,000 per year. The cost of capital of Unnat Ltd. is 12% and acceptable level of risk is 20%.

You are required to measure the sensitivity of the project's net present value to a change in the following project variables:

(a) sale price;

(b) sales volume;

(c) variable cost;

(d) On further investigation it is found that there is a significant chance that the expected sales volume of 2,00,000 units per year will not be achieved. The sales manager of Unnat Ltd. suggests that sales volumes could depend on expected economic states which could be assigned the following probabilities:

| State of Economy | Annual Sales (in Units) | Prob. |
|---|---|---|
| Poor | 1,75000 | 0·30 |
| Normal | 2,00,000 | 0·60 |
| Good | 2,25,000 | 0·10 |

Calculate expected net present value of the project and give your decision whether company should accept the project or not.

## Solution — ICAI

**13.** Calculation of NPV

= - ₹ 50,00,000 + [2,00,000 (₹ 30 – ₹ 16.50) – ₹ 10,00,000] PVIAF (12%,5)

= - ₹ 50,00,000 + [2,00,000 (₹ 13.50) – ₹ 10,00,000] 3.605

= - ₹ 50,00,000 + [₹ 27,00,000 – ₹ 10,00,000] 3.605

= - ₹ 50,00,000 + ₹ 61,28,500 = ₹ 11,28,500

Measurement of Sensitivity Analysis

**(a) Sales Price:-**

Let the sale price/Unit be S so that the project would break even with 0 NPV.

∴ ₹ 50,00,000 = [2,00,000 (S – ₹ 16.50) – ₹ 10,00,000] PVIAF (12%,5)

₹ 50,00,000 = [2,00,000S – ₹ 33,00,000 – ₹ 10,00,000] 3.605

₹ 50,00,000 = [2,00,000S – ₹ 43,00,000] 3.605

₹ 13,86,963 = 2,00,000S – ₹ 43,00,000

₹ 56,86,963 = 2,00,000S

S = ₹ 28.43 which represents a fall of (30 - 28.43)/30 or 0.0523 or 5.23%

**(b) Sales volume:-**

Let V be the sale volume so that the project would break even with 0 NPV.

₹ 50,00,000 = [V (₹ 30 – ₹ 16.50) – ₹ 10,00,000] PVIAF (12%,5)

₹ 50,00,000 = [V (₹ 13.50) – ₹ 10,00,000] PVIAF (12%,5)

₹ 50,00,000 = [₹ 13.50V – ₹ 10,00,000] 3.605

₹ 13,86,963 = ₹ 13.50V – ₹ 10,00,000

₹ 23,86,963 = ₹ 13.50V

V = 1,76,812 which represents a fall of (2,00,000 - 1,76,812)/2,00,000 or 0.1159 or 11.59%

**(c) Variable Cost:**

Let the variable cost be V so that the project would break even with 0 NPV.

∴ ₹ 50,00,000 = [2,00,000(₹ 30 – V) – ₹ 10,00,000] PVIAF(12%,5)

₹ 50,00,000 = [₹ 60,00,000 – 2,00,000 V – ₹ 10,00,000] 3.605

₹ 50,00,000 = [₹ 50,00,000 – 2,00,000 V] 3.605

₹ 13,86,963 = ₹ 50,00,000 – 2,00,000 V

₹ 36,13,037 = 2,00,000V

V = ₹ 18.07 which represents a fall of (18.07 – 16.50)/16.50 or 0.0951 or 9.51%

**(d) Expected Net Present Value**

(1,75,000 X 0.30) + (2,00,000 X 0.60) + (2,25,000 X 0.10) =1,95,000

NPV = [1,95,000 X ₹ 13.50 – ₹ 10,00,000] 3.605 – ₹ 50,00,000 = ₹ 8,85,163

Further NPV in worst and best cases will be as follows:

Worst Case:

[1,75,000 X ₹ 13.50 – ₹ 10,00,000] 3.605 – ₹ 50,00,000 = - ₹ 88,188

Best Case:

[2,25,000 X ₹ 13.50 – ₹ 10,00,000] 3.605 – ₹ 50,00,000 = ₹ 23,45,188

Thus, there are 30% chances that the rise will be a negative NPV and 70% chances of positive NPV. Since acceptable level of risk of Unnat Ltd. is 20% and there are 30% chances of negative NPV hence project should not be accepted.

## Practical Q14

14. The Textile Manufacturing Company Ltd., is considering one of two mutually exclusive proposals, Projects M and N, which require cash outlays of ₹ 8,50,000 and ₹ 8,25,000 respectively. The certainty-equivalent (C.E) approach is used in incorporating risk in capital budgeting decisions. The current yield on government bonds is 6% and this is used as the risk free rate. The expected net cash flows and their certainty equivalents are as follows:

| Project M | | | Project N | |
|---|---|---|---|---|
| Year-end | Cash Flow ₹ | C.E. | Cash Flow ₹ | C.E. |
| 1 | 4,50,000 | 0.8 | 4,50,000 | 0.9 |
| 2 | 5,00,000 | 0.7 | 4,50,000 | 0.8 |
| 3 | 5,00,000 | 0.5 | 5,00,000 | 0.7 |

Present value factors of ₹ 1 discounted at 6% at the end of year 1, 2 and 3 are 0.943, 0.890 and 0.840 respectively.

Required:

(i) Which project should be accepted?

(ii) If risk adjusted discount rate method is used, which project would be appraised with a higher rate and why?

## Solution — ICAI

**14. (i) Statement Showing the Net Present Value of Project M**

| Year end | Cash Flow (₹) (a) | C.E. (b) | Adjusted Cash flow (₹) (c) = (a) × (b) | Present value factor at 6% (d) | Total Present value (₹) (e) = (c) × (d) |
|---|---|---|---|---|---|
| 1 | 4,50,000 | 0.8 | 3,60,000 | 0.943 | 3,39,480 |
| 2 | 5,00,000 | 0.7 | 3,50,000 | 0.890 | 3,11,500 |
| 3 | 5,00,000 | 0.5 | 2,50,000 | 0.840 | 2,10,000 |
| | | | | | 8,60,980 |
| Less: Initial Investment | | | | | 8,50,000 |
| Net Present Value | | | | | 10,980 |

**Statement Showing the Net Present Value of Project N**

| Year end | Cash Flow (₹) (a) | C.E. (b) | Adjusted Cash flow (₹) (c) = (a) × (b) | Present value factor (d) | Total Present value (₹) (e) = (c) × (d) |
|---|---|---|---|---|---|
| 1 | 4,50,000 | 0.9 | 4,05,000 | 0.943 | 3,81,915 |
| 2 | 4,50,000 | 0.8 | 3,60,000 | 0.890 | 3,20,400 |
| 3 | 5,00,000 | 0.7 | 3,50,000 | 0.840 | 2,94,000 |
| | | | | | 9,96,315 |
| Less: Initial Investment | | | | | 8,25,000 |
| Net Present Value | | | | | 1,71,315 |

**Decision:** Since the net present value of Project N is higher, so the project N should be accepted.

**(ii)** Certainty - Equivalent (C.E.) Co-efficient of Project M (2.0) is lower than Project N (2.4). This means Project M is riskier than Project N as "higher the riskiness of a cash flow, the lower will be the CE factor". If risk adjusted discount rate (RADR) method is used, Project M would be analysed with a higher rate.

---

RADR is based on the premise that riskiness of a proposal may be taken care of, by adjusting the discount rate. The cash flows from a more risky proposal should be discounted at a relatively higher discount rate as compared to other proposals whose cash flows are less risky. Any investor is basically risk averse. However, he may be ready to take risk provided he is rewarded for undertaking risk by higher returns. So, more risky the investment is, the greater would be the expected return. The expected return is expressed in terms of discount rate which is also the minimum required rate of return generated by a proposal if it is to be accepted. Therefore, there is a positive correlation between risk of a proposal and the discount rate.

## Practical Q15

15. Determine the risk adjusted net present value of the following projects:

| | X | Y | Z |
|---|---|---|---|
| Net cash outlays (₹) | 2,10,000 | 1,20,000 | 1,00,000 |
| Project life | 5 years | 5 years | 5 years |
| Annual Cash inflow (₹) | 70,000 | 42,000 | 30,000 |
| Coefficient of variation | 1.2 | 0.8 | 0.4 |

The Company selects the risk-adjusted rate of discount on the basis of the coefficient of variation:

| Coefficient of Variation | Risk-Adjusted Rate of Return | P.V. Factor 1 to 5 years At risk adjusted rate of discount |
|---|---|---|
| 0.0 | 10% | 3.791 |
| 0.4 | 12% | 3.605 |
| 0.8 | 14% | 3.433 |
| 1.2 | 16% | 3.274 |
| 1.6 | 18% | 3.127 |
| 2.0 | 22% | 2.864 |
| More than 2.0 | 25% | 2.689 |

## Solution — ICAI

15. **Statement showing the determination of the risk adjusted net present value**

| *Projects* | *Net cash outlays* | *Coefficient of variation* | *Risk adjusted discount rate* | *Annual cash inflow* | *PV factor 1-5 years* | *Discounted cash inflow* | *Net present value* |
|---|---|---|---|---|---|---|---|
| | ₹ | | | ₹ | | ₹ | ₹ |
| *(i)* | *(ii)* | *(iii)* | *(iv)* | *(v)* | *(vi)* | *(vii) = (v) × (vi)* | *(viii) = (vii) – (ii)* |
| X | 2,10,000 | 1.20 | 16% | 70,000 | 3.274 | 2,29,180 | 19,180 |
| Y | 1,20,000 | 0.80 | 14% | 42,000 | 3.433 | 1,44,186 | 24,186 |
| Z | 1,00,000 | 0.40 | 12% | 30,000 | 3.605 | 1,08,150 | 8,150 |

## Practical Q16

16. New Projects Ltd. is evaluating 3 projects, P-I, P-II, P-III. Following information is available in respect of these projects:

| | P-I | P-II | P-III |
|---|---|---|---|
| Cost | ₹ 15,00,000 | ₹ 11,00,000 | ₹ 19,00,000 |
| Inflows-Year 1 | 6,00,000 | 6,00,000 | 4,00,000 |
| Year 2 | 6,00,000 | 4,00,000 | 6,00,000 |
| Year 3 | 6,00,000 | 5,00,000 | 8,00,000 |
| Year 4 | 6,00,000 | 2,00,000 | 12,00,000 |
| Risk Index | 1.80 | 1.00 | 0.60 |

Minimum required rate of return of the firm is 15% and applicable tax rate is 40%. The risk free interest rate is 10%.

Required:

(i) Find out the risk-adjusted discount rate (RADR) for these projects.

(ii) Which project is the best?

## Solution — ICAI

**16. (i)** The risk free rate of interest and risk factor for each of the projects are given. The risk adjusted discount rate (RADR) for different projects can be found on the basis of CAPM as follows:

Required Rate of Return = $I_{Rf}$ + ($k_o$ - $I_{RF}$ ) Risk Factor

For P-I : RADR = 0.10 + (0.15 – 0.10 ) 1.80 = 19%

For P-II : RADR = 0.10 + (0.15 – 0.10 ) 1.00 = 15 %

For P-III : RADR = 0.10 + (0.15 – 0.10) 0.60 = 13 %

**(ii)** The three projects can now be evaluated at 19%, 15% and 13% discount rate as follows:

**Project P-I**

| | |
|---|---|
| Annual Inflows | ₹ 6,00,000 |
| PVAF (19 %, 4) | 2.639 |
| PV of Inflows (₹ 6,00,000 x 2.639) | ₹ 15,83,400 |

| | |
|---|---|
| *Less*: Cost of Investment | ₹ 15,00,000 |
| Net Present Value | ₹ 83,400 |

**Project P-II**

| Year | Cash Inflow (₹) | PVF (15%,n) | PV (₹) |
|---|---|---|---|
| 1 | 6,00,000 | 0.870 | 5,22,000 |
| 2 | 4,00,000 | 0.756 | 3,02,400 |
| 3 | 5,00,000 | 0.658 | 3,29,000 |
| 4 | 2,00,000 | 0.572 | 1,14,400 |
| Total Present Value | | | 12,67,800 |
| Less: Cost of Investment | | | 11,00,000 |
| Net Present Value | | | **1,67,800** |

**Project P-III**

| Year | Cash Inflow (₹) | PVF (13%,n) | PV (₹) |
|---|---|---|---|
| 1 | 4,00,000 | 0.885 | 3,54,000 |
| 2 | 6,00,000 | 0.783 | 4,69,800 |
| 3 | 8,00,000 | 0.693 | 5,54,400 |
| 4 | 12,00,000 | 0.613 | 7,35,600 |
| Total Present Value | | | 21,13,800 |
| *Less*: Cost of Investment | | | 19,00,000 |
| Net Present Value | | | **2,13,800** |

Project P-III has highest NPV. So, it should be accepted by the firm

## Practical Q17

17. A firm has projected the following cash flows from a project under evaluation:

| Year | ₹ lakhs |
|---|---|
| 0 | (70) |
| 1 | 30 |
| 2 | 40 |
| 3 | 30 |

The above cash flows have been made at expected prices after recognizing inflation. The firm's cost of capital is 10%. The expected annual rate of inflation is 5%.

Show how the viability of the project is to be evaluated.

## Solution — ICAI

**17.** It is stated that the cash flows have been adjusted for inflation; hence they are "nominal". The cost of capital or discount rate is "real". In order to be compatible, the cash flows should be converted into "real flow". This is done as below:

| Year | Nominal cash flows | Adjusted Inflation* factor | Real cash flows | PVF @ 10% | PV of cash flows |
|---|---|---|---|---|---|
| 0 | (70) | – | (70) | 1.000 | (70) |
| 1 | 30 | 0.952 | 28.56 | 0.909 | 25.96 |
| 2 | 40 | 0.907 | 36.28 | 0.826 | 29.97 |

| | | | | | |
|---|---|---|---|---|---|
| 3 | 30 | 0.864 | 25.92 | 0.751 | 19.47 |
| | | | | Total | 75.40 |
| *Less:* Cash out flow | | | | | 70.00 |
| NPV (+) | | | | | 5.40 |

* $1/1.05$; $1/(1.05)^2$; $1/(1.05)^3$;

*Advise:* With positive NPV, the project is financially viable.

Alternatively, instead of converting cash flows into real terms, the discount rate can be converted into nominal rate. Result will be the same.

An alternative solution is presented herewith

**Alternative solution:**

| Year | Nominal cash flows | PVF @ 15.50% adjusted by the inflation factor i.e. 5%* | PV of cash flows |
|---|---|---|---|
| 1 | 30 | 0.866 | 25.98 |
| 2 | 40 | 0.749 | 29.96 |
| 3 | 30 | 0.649 | 19.47 |
| | | Cash inflow | 75.41 |
| | | *Less:* Cash out flow | 70.00 |
| | | Net present value | 5.41 |

* $\frac{0.909}{1.05}=0.866, \frac{0.826}{1.1025}=0.749, \frac{0.751}{1.1576}=0.649$

Advise: With positive NPV, the project is financially viable.

## Practical Q18

18. Shashi Co. Ltd has projected the following cash flows from a project under evaluation:

| Year | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| ₹ (in lakhs) | (72) | 30 | 40 | 30 |

The above cash flows have been made at expected prices after recognizing inflation. The firm's cost of capital is 10%. The expected annual rate of inflation is 5%. Show how the viability of the project is to be evaluated. PVF at 10% for 1-3 years are 0.909, 0.826 and 0.751.

## Solution — ICAI

**18.** Here the given cash flows have to be adjusted for inflation. Alternatively, the discount rate can be converted into nominal rate, as follows:-

Year 1 = $\frac{0.909}{1.05} = 0.866$; Year 2 = $\frac{0.826}{(1.05)^2}$ or $\frac{0.826}{1.1025} = 0.749$

Year 3 = $\frac{0.751}{(1.05)^3} = \frac{0.751}{1.1576} = 0.649$

| Year | Nominal Cash Flows (₹ in lakhs) | Adjusted PVF as above | PV of Cash Flows (₹ in lakhs) |
|---|---|---|---|
| 1 | 30 | 0.866 | 25.98 |
| 2 | 40 | 0.749 | 29.96 |
| 3 | 30 | 0.649 | 19.47 |
| | Cash Inflow | | 75.41 |
| | *Less*: Cash Outflow | | 72.00 |
| | Net Present Value | | 3.41 |

With positive NPV, the project is financially viable.

**Alternative Solution**

Assumption: The cost of capital given in the question is "Real'.

Nominal cost of capital = (1.10) (1.05) -1 = 0.155 =15.50%

DCF Analysis of the project

(₹ Lakhs)

| | Period | PVF @15.50% | CF | PV |
|---|---|---|---|---|
| Investment | 0 | 1 | -72 | -72.00 |
| Operation | 1 | 0.866 | 30 | +25.98 |
| ---do--- | 2 | 0.750 | 40 | +30.00 |
| ---do--- | 3 | 0.649 | 30 | +19.47 |
| NPV | | | | +3.45 |

The proposal may be accepted as the NPV is positive.

## Practical Q19

19. KLM Ltd. requires ₹ 15,00,000 for a new project.

Useful life of project is 3 years.

Salvage value - NIL.

Depreciation is ₹ 5,00,000 p.a.

Given below are projected revenues and costs (excluding depreciation) ignoring inflation:

| Year → | 1 | 2 | 3 |
|---|---|---|---|
| Revenues in ₹ | 10,00,000 | 13,00,000 | 14,00,000 |
| Costs in ₹ | 5,00,000 | 6,00,000 | 6,50,000 |

Applicable tax rate is 35%. Assume nominal cost of capital to be 14% (after tax). The inflation rates for revenues and costs are as under:

| Year | Revenues % | Costs % |
|---|---|---|
| 1 | 9 | 10 |
| 2 | 8 | 9 |
| 3 | 6 | 7 |

PVF at 14%, for 3 years = 0.877, 0.769 and 0.675

Show amount to the nearest rupee in calculations.

You are required to calculate net present value of the project.

## Solution — ICAI

**19. (i) Inflation adjusted Revenues**

| Year | Revenues (₹) | Revenues (Inflation Adjusted) (₹) |
|---|---|---|
| 1 | 10,00,000 | 10,00,000(1.09) = 10,90,000 |
| 2 | 13,00,000 | 13,00,000(1.09) (1.08) = 15,30,360 |
| 3 | 14,00,000 | 14,00,000(1.09) (1.08)(1.06) = 17,46,965 |

(ii) **Inflation adjusted Costs**

| Year | Costs (₹) | Costs (Inflation Adjusted) (₹) |
|---|---|---|
| 1 | 5,00,000 | 5,00,000(1.10) = 5,50,000 |
| 2 | 6,00,000 | 6,00,000(1.10)(1.09) = 7,19,400 |
| 3 | 6,50,000 | 6,50,000(1.10)(1.09)(1.07) = 8,33,905 |

(iii) Tax Benefit on Depreciation = ₹ 5,00,000 x 0.35 = ₹ 1,75,000

(iv) **Net Profit after Tax**

| Year | Revenues (Inflation Adjusted) (₹) (1) | Costs (Inflation Adjusted) (₹) (2) | Net Profit (₹) (3) =(1) -(2) | Tax (₹) (4) = 35% of (3) | Profit after Tax (₹) (3) - (4) |
|---|---|---|---|---|---|
| 1 | 10,90,000 | 5,50,000 | 5,40,000 | 1,89,000 | 3,51,000 |
| 2 | 15,30,360 | 7,19,400 | 8,10,960 | 2,83,836 | 5,27,124 |
| 3 | 17,46,965 | 8,33,905 | 9,13,060 | 3,19,571 | 5,93,489 |

(v) **Present Value of Cash Inflows**

| Year | Net Profit after tax (₹) | Tax Benefit on Depreciation (₹) | Cash Inflow (₹) | PVF@ 14% | PV (₹) |
|---|---|---|---|---|---|
| 1 | 3,51,000 | 1,75,000 | 5,26,000 | 0.877 | 4,61,302 |
| 2 | 5,27,124 | 1,75,000 | 7,02,124 | 0.769 | 5,39,933 |
| 3 | 5,93,489 | 1,75,000 | 7,68,489 | 0.675 | 5,18,730 |
| | | | | | 15,19,965 |

NPV = ₹ 15,19,965 – ₹ 15,00,000 = ₹ 19,965

## Practical Q20

20. A firm has an investment proposal, requiring an outlay of ₹ 80,000. The investment proposal is expected to have two years economic life with no salvage value. In year 1, there is a 0.4 probability that cash inflow after tax will be ₹ 50,000 and 0.6 probability that cash inflow after tax will be ₹ 60,000. The probability assigned to cash inflow after tax for the year 2 is as follows:

| The cash inflow year 1 | ₹ 50,000 | | ₹ 60,000 | |
|---|---|---|---|---|
| The cash inflow year 2 | Probability | | Probability | |
| | ₹ 24,000 | 0.2 | ₹ 40,000 | 0.4 |
| | ₹ 32,000 | 0.3 | ₹ 50,000 | 0.5 |
| | ₹ 44,000 | 0.5 | ₹ 60,000 | 0.1 |

The firm uses a 10% discount rate for this type of investment.

Required:

(i) Construct a decision tree for the proposed investment project and calculate the expected net present value (NPV).

(ii) What net present value will the project yield, if worst outcome is realized? What is the probability of occurrence of this NPV?

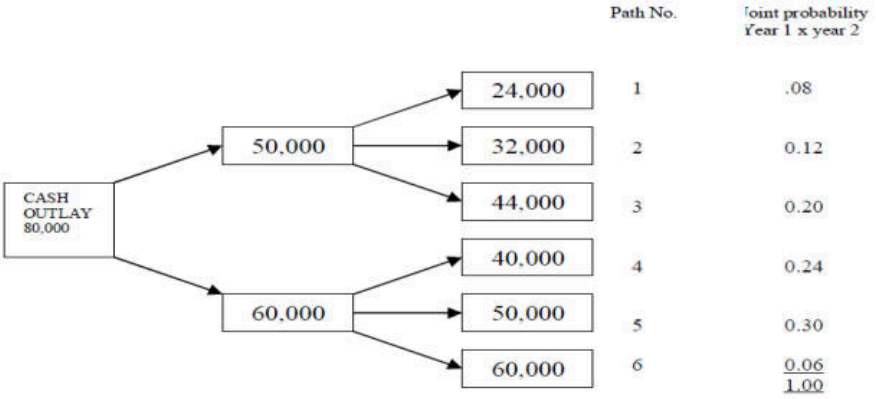
(iii) What will be the best outcome and the probability of that occurrence?

(iv) Will the project be accepted?

(Note: 10% discount factor 1 year 0.909; 2 year 0.826)

## Solution — ICAI

**20. (i)** The decision tree diagram is presented in the chart, identifying various paths and outcomes, and the computation of various paths/outcomes and NPV of each path are presented in the following tables:




The Net Present Value (NPV) of each path at 10% discount rate is given below:

| Path | Year 1 Cash Flows (₹) | Year 2 Cash Flows (₹) | Total Cash Inflows (PV) (₹) | Cash Inflows NPV (₹) | (₹) |
|---|---|---|---|---|---|
| 1 | 50,000×.909 = 45,450 | 24,000×.826 = 19,824 | 65,274 | 80,000 | (–) 14,726 |
| 2 | 45,450 | 32,000×.826 = 26,432 | 71,882 | 80,000 | (–) 8,118 |
| 3 | 45,450 | 44,000×.826 = 36,344 | 81,794 | 80,000 | 1,794 |
| 4 | 60,000×.909 = 54,540 | 40,000×.826 = 33,040 | 87,580 | 80,000 | 7,580 |
| 5 | 54,540 | 50,000×.826 = 41,300 | 95,840 | 80,000 | 15,840 |
| 6 | 54,540 | 60,000×.826 = 49,560 | 1,04,100 | 80,000 | 24,100 |

**Statement showing Expected Net Present Value**

₹

| z | NPV (₹) | Joint Probability | Expected NPV |
|---|---|---|---|
| 1 | –14,726 | 0.08 | –1,178.08 |
| 2 | –8,118 | 0.12 | –974.16 |
| 3 | 1,794 | 0.20 | 358.80 |
| 4 | 7,580 | 0.24 | 1,819.20 |
| 5 | 15,840 | 0.30 | 4,752.00 |
| 6 | 24,100 | 0.06 | 1,446.00 |
| | | | 6,223.76 |

**(ii)** If the worst outcome is realized the project will yield NPV of – ₹ 14,726. The probability of occurrence of this NPV is 8% and a loss of ₹ 1,178 (path 1).

**(iii)** The best outcome will be path 6 when the NPV is at ₹ 24,100. The probability of occurrence of this NPV is 6% and a expected profit of ₹ 1,446.

**(iv)** The project should be accepted because the expected NPV is positive at ₹ 6,223.76 based on joint probability.

## Practical Q21

21. Jumble Consultancy Group has determined relative utilities of cash flows of two forthcoming projects of its client company as follows:

| Cash Flow in ₹ | -15000 | -10000 | -4000 | 0 | 15000 | 10000 | 5000 | 1000 |
|---|---|---|---|---|---|---|---|---|
| Utilities | -100 | -60 | -3 | 0 | 40 | 30 | 20 | 10 |

The distribution of cash flows of project A and Project B are as follows:

| **Project A** | | | | | |
|---|---|---|---|---|---|
| Cash Flow (₹) | -15000 | - 10000 | 15000 | 10000 | 5000 |
| Probability | 0.10 | 0.20 | 0.40 | 0.20 | 0.10 |
| **Project B** | | | | | |
| Cash Flow (₹) | - 10000 | -4000 | 15000 | 5000 | 10000 |
| Probability | 0.10 | 0.15 | 0.40 | 0.25 | 0.10 |

Which project should be selected and why ?

## Solution — ICAI

**21.** Evaluation of project utilizes of Project A and Project B

| | Project A | | |
|---|---|---|---|
| **Cash flow (in ₹)** | **Probability** | **Utility** | **Utility value** |
| -15,000 | 0.10 | -100 | -10 |
| -10,000 | 0.20 | -60 | -12 |
| 15,000 | 0.40 | 40 | 16 |
| 10,000 | 0.20 | 30 | 6 |
| 5,000 | 0.10 | 20 | 2 |
| | | | 2 |

| **Cash flow (in ₹)** | Project B | | |
|---|---|---|---|
| | **Probability** | **Utility** | **Utility value** |
| -10,000 | 0.10 | -60 | -6 |
| -4,000 | 0.15 | -3 | -0.45 |
| 15,000 | 0.40 | 40 | 16 |
| 5,000 | 0.25 | 20 | 5 |
| 10,000 | 0.10 | 30 | 3 |
| | | | 17.55 |

Project B should be selected as its expected utility is more.

## Practical Q22

22. A & Co. is contemplating whether to replace an existing machine or to spend money on overhauling it. A & Co. currently pays no taxes. The replacement machine costs ₹ 90,000 now and requires maintenance of ₹ 10,000 at the end of every year for eight years. At the end of eight years it would have a salvage value of ₹ 20,000 and would be sold. The existing machine requires increasing amounts of maintenance each year and its salvage value falls each year as follows:

| Year | Maintenance (₹) | Salvage (₹) |
|---|---|---|
| Present | 0 | 40,000 |
| 1 | 10,000 | 25,000 |
| 2 | 20,000 | 15,000 |
| 3 | 30,000 | 10,000 |
| 4 | 40,000 | 0 |

The opportunity cost of capital for A & Co. is 15%.

Required:

When should the company replace the machine?

(Notes: Present value of an annuity of Re. 1 per period for 8 years at interest rate of 15% : 4.4873; present value of Re. 1 to be received after 8 years at interest rate of 15% : 0.3269).

## Solution — ICAI

22. **A & Co.**

**Equivalent cost of (EAC) of new machine**

| | | ₹ |
|---|---|---|
| (i) | Cost of new machine now | 90,000 |
| | *Add:* PV of annual repairs @ ₹ 10,000 per annum for 8 years (₹ 10,000 × 4.4873) | 44,873 |
| | | 1,34,873 |
| | *Less:* PV of salvage value at the end of 8 years (₹20,000×0.3269) | 6,538 |
| | | 1,28,335 |
| | Equivalent annual cost (EAC) (₹ 1,28,335/4.4873) | 28,600 |

**PV of cost of replacing the old machine in each of 4 years with new machine**

| Scenario | Year | Cash Flow (₹) | PV @ 15% | PV (₹) |
|---|---|---|---|---|
| Replace Immediately | 0 | (28,600) | 1.00 | (28,600) |
| | | 40,000 | 1.00 | 40,000 |
| | | | | 11,400 |
| Replace in one year | 1 | (28,600) | 0.870 | (24,882) |
| | 1 | (10,000) | 0.870 | (8,700) |
| | 1 | 25,000 | 0.870 | 21,750 |
| | | | | (11,832) |
| Replace in two years | 1 | (10,000) | 0.870 | (8,700) |
| | 2 | (28,600) | 0.756 | (21,622) |
| | 2 | (20,000) | 0.756 | (15,120) |
| | 2 | 15,000 | 0.756 | 11,340 |
| | | | | (34,102) |
| Replace in three years | 1 | (10,000) | 0.870 | (8,700) |
| | 2 | (20,000) | 0.756 | (15,120) |
| | 3 | (28,600) | 0.658 | (18,819) |
| | 3 | (30,000) | 0.658 | (19,740) |
| | 3 | 10,000 | 0.658 | 6,580 |
| | | | | (55,799) |
| Replace in four years | 1 | (10,000) | 0.870 | (8,700) |
| | 2 | (20,000) | 0.756 | (15,120) |
| | 3 | (30,000) | 0.658 | (19,740) |
| | 4 | (28,600) | 0.572 | (16,359) |
| | 4 | (40,000) | 0.572 | (22,880) |
| | | | | (82,799) |

**Advice:** The company should replace the old machine immediately because the PV of cost of replacing the old machine with new machine is least.

**Alternatively, optimal replacement period can also be computed using the following table:**

| Scenario | Year | Cashflow | PV at 15% | PV |
|---|---|---|---|---|
| Replace immediately | 0 | (40,000) | 1 | (40,000) |
| | 1 to 4 | 28,600 | 2.855 | 81,652 |
| | | | | **41,652** |
| Replace after 1 year | 1 | 10,000 | 0.870 | 8,696 |
| | 1 | (25,000) | 0.870 | (21,739) |
| | 2 to 4 | 28,600 | 1.985 | 56,783 |
| | | | | **43,739** |
| Replace after 2 years | 1 | 10,000 | 0.870 | 8,696 |
| | 2 | 20,000 | 0.756 | 15,123 |
| | 2 | (15,000) | 0.756 | (11,342) |
| | 3 and 4 | 28,600 | 1.229 | 35,157 |
| | | | | **47,633** |
| Replace after 3 years | 1 | 10,000 | 0.870 | 8,696 |
| | 2 | 20,000 | 0.756 | 15,123 |
| | 3 | 30,000 | 0.658 | 19,725 |
| | 3 | (10,000) | 0.658 | (6,575) |
| | 4 | 28,600 | 0.572 | 16,352 |
| | | | | **53,321** |
| Replace after 4 years | 1 | 10,000 | 0.870 | 8,696 |
| | 2 | 20,000 | 0.756 | 15,123 |
| | 3 | 30,000 | 0.658 | 19,725 |
| | 4 | 40,000 | 0.572 | 22,870 |
| | | | | **66,414** |

## Practical Q23

23. A company has an old machine having book value zero – which can be sold for ₹ 50,000. The company is thinking to choose one from following two alternatives:

(i) To incur additional cost of ₹ 10,00,000 to upgrade the old existing machine.

(ii) To replace old machine with a new machine costing ₹ 20,00,000 plus installation cost ₹ 50,000.

Both above proposals envisage useful life to be five years with salvage value to be nil.

The expected after tax profits for the above three alternatives are as under :

| Year | Old existing Machine (₹) | Upgraded Machine (₹) | New Machine (₹) |
|---|---|---|---|
| 1 | 5,00,000 | 5,50,000 | 6,00,000 |
| 2 | 5,40,000 | 5,90,000 | 6,40,000 |
| 3 | 5,80,000 | 6,10,000 | 6,90,000 |
| 4 | 6,20,000 | 6,50,000 | 7,40,000 |
| 5 | 6,60,000 | 7,00,000 | 8,00,000 |

The tax rate is 40 per cent.

The company follows straight line method of depreciation. Assume cost of capital to be 15 per cent.

P.V.F. of 15%, 5 = 0.870, 0.756, 0.658, 0.572 and 0.497. You are required to advise the company as to which alternative is to be adopted.

## Solution — ICAI

**23.**

| (A) | Cash Outflow | | ₹ |
|---|---|---|---|
| | (i) | In case machine is upgraded: | |
| | | Upgradation Cost | 10,00,000 |
| | (ii) | In case new machine installed: | |
| | | Cost | 20,00,000 |
| | | *Add*: Installation cost | 50,000 |
| | | Total Cost | 20,50,000 |
| | | *Less*: Disposal of old machine | |
| | | ₹ 50,000 – 40% tax | 30,000 |
| | | Total Cash Outflow | 20,20,000 |

**Working Note:**

(i) Depreciation – in case machine is upgraded

₹ 10,00,000 ÷ 5 = ₹ 2,00,000

(ii) Depreciation – in case new machine is installed

₹ 20,50,000 ÷ 5 = ₹ 4,10,000

(iii) Old existing machine – Book Value is zero. So, no depreciation.

**(B) Cash Inflows after Taxes (CFAT)**

| Year | Old Existing Machine | Upgraded Machine | | | |
|---|---|---|---|---|---|
| | (i) EAT/CFAT ₹ | (ii) EAT ₹ | (iii) DEP ₹ | (iv) CFAT ₹ | = (iv)-(i) Incremental CFAT ₹ |
| 1 | 5,00,000 | 5,50,000 | 2,00,000 | 7,50,000 | 2,50,000 |
| 2 | 5,40,000 | 5,90,000 | 2,00,000 | 7,90,000 | 2,50,000 |
| 3 | 5,80,000 | 6,10,000 | 2,00,000 | 8,10,000 | 2,30,000 |
| 4 | 6,20,000 | 6,50,000 | 2,00,000 | 8,50,000 | 2,30,000 |
| 5 | 6,60,000 | 7,00,000 | 2,00,000 | 9,00,000 | 2,40,000 |

**Cash Inflow after Taxes (CFAT)**

| Year | | New Machine | | |
|---|---|---|---|---|
| | (vi) EAT ₹ | (vii) DEP ₹ | (viii) CFAT ₹ | (ix) = (viii) – (i) Incremental CFAT (₹ ) |
| 1 | 6,00,000 | 4,10,000 | 10,10,000 | 5,10,000 |
| 2 | 6,40,000 | 4,10,000 | 10,50,000 | 5,10,000 |
| 3 | 6,90,000 | 4,10,000 | 11,00,000 | 5,20,000 |
| 4 | 7,40,000 | 4,10,000 | 11,50,000 | 5,30,000 |
| 5 | 8,00,000 | 4,10,000 | 12,10,000 | 5,50,000 |

**P.V. AT 15% - 5 Years – on Incremental CFAT**

| Year | Upgraded Machine | | | New Machine | | |
|---|---|---|---|---|---|---|
| | Incremental CFAT ₹ | PVF | Total P.V. ₹ | Incremental CFAT | PVF | Total PV ₹ |
| 1 | 2,50,000 | 0.870 | 2,17,500 | 5,10,000 | 0.870 | 4,43,700 |
| 2 | 2,50,000 | 0.756 | 1,89,000 | 5,10,000 | 0.756 | 3,85,560 |
| 3 | 2,30,000 | 0.658 | 1,51,340 | 5,20,000 | 0.658 | 3,42,160 |
| 4 | 2,30,000 | 0.572 | 1,31,560 | 5,30,000 | 0.572 | 3,03,160 |
| 5. | 2,40,000 | 0.497 | 1,19,280 | 5,50,000 | 0.497 | 2,73,350 |
| Total P.V. of CFAT | | | 8,08,680 | | | 17,47,930 |
| *Less*: Cash Outflows | | | 10,00,000 | | | 20,20,000* |
| N.P.V. = | | | -1,91,320 | | | - 2,72,070 |

| | |
|---|---|
| *Acquisition Cost (including installation cost) | ₹ 20,50,000 |
| *Less*: Salvage Value of existing machine net of Tax | ₹ 30,000 |
| | ₹ 20,20,000 |

As the NPV in both the new (alternative) proposals is negative, the company should continue with the existing old Machine.

## Practical Q24

24. Company X is forced to choose between two machines A and B. The two machines are designed differently but have identical capacity and do exactly the same job. Machine A costs ₹ 1,50,000 and will last for 3 years. It costs ₹ 40,000 per year to run. Machine B is an 'economy' model costing only ₹ 1,00,000, but will last only for 2 years, and costs ₹ 60,000 per year to run. These are real cash flows. The costs are forecasted in rupees of constant purchasing power. Ignore tax. Opportunity cost of capital is 10 per cent. Which machine company X should buy?

## Solution — ICAI

**24. Statement showing the evaluation of two machines**

| Machines | A | B |
|---|---|---|
| Purchase cost (₹): (i) | 1,50,000 | 1,00,000 |
| Life of machines (years) | 3 | 2 |
| Running cost of machine per year (₹): (ii) | 40,000 | 60,000 |
| Cumulative present value factor for 1-3 years @ 10% (iii) | 2.486 | – |
| Cumulative present value factor for 1-2 years @ 10% (iv) | – | 1.735 |
| Present value of running cost of machines (₹): (v) | 99,440 [(ii) × (iii)] | 1,04,100 [(ii) × (iv)] |
| Cash outflow of machines (₹): (vi) = (i) + (v) | 2,49,440 | 2,04,100 |
| Equivalent present value of annual cash outflow | 1,00,338 [(vi) ÷ (iii)] | 1,17,637 [(vi) ÷ (iv)] |

**Decision:** Company X should buy machine A since its equivalent cash outflow is less than machine B.

## Practical Q25

25. Company Y is operating an elderly machine that is expected to produce a net cash inflow of ₹ 40,000 in the coming year and ₹ 40,000 next year. Current salvage value is ₹ 80,000 and next year's value is ₹ 70,000. The machine can be replaced now with a new machine, which costs ₹ 1,50,000, but is much more efficient and will provide a cash inflow of ₹ 80,000 a year for 3 years. Company Y wants to know whether it should replace the equipment now or wait a year with the clear understanding that the new machine is the best of the available alternatives and that it in turn be replaced at the optimal point. Ignore tax. Take opportunity cost of capital as 10 per cent. Advise with reasons.

## Solution — ICAI

**25. Statement showing present value of cash inflow of new machine when it replaces elderly machine now**

**NPV of New Machine**

| | |
|---|---|
| PV of Cash Inflow (80000 x 2.486) | ₹ 1,98,880 |
| Less: Purchase Cost of New Machine | ₹ 1,50,000 |
| | ₹ 48,880 |

Since NPV of New Machine is positive, it should be purchased.

**Timing Decision**

Replace Now

| | |
|---|---|
| Current Realizable Value | ₹ 80,000 |
| NPV of New Machine | ₹ 48,880 |
| Total NPV | ₹ 1,28,880 |

Replace after 1 Year

| | |
|---|---|
| Cash Inflow for Year 1 | ₹ 40000 |
| Realisable Value of Old Machine | ₹ 70000 |

| | |
|---|---|
| NPV of New Machine | ₹ 48,880 |
| Total NPV after 1 Year | ₹ 1,58,880 |
| PV of Total NPV (158880/1.1) | ₹ 1,44,436 |

**Advise:** Since Total NPV is higher in case of Replacement after one year Machine should be replaced after 1 year.

## Practical Q26

26. A machine used on a production line must be replaced at least every four years. Costs incurred to run the machine according to its age are:

| Age of the Machine (years) | | | | | |
|---|---|---|---|---|---|
| | 0 | 1 | 2 | 3 | 4 |
| Purchase price (in ₹) | 60,000 | | | | |
| Maintenance (in ₹) | | 16,000 | 18,000 | 20,000 | 20,000 |
| Repair (in ₹) | | 0 | 4,000 | 8,000 | 16,000 |
| Scrap Value (in ₹) | | 32,000 | 24,000 | 16,000 | 8,000 |

Future replacement will be with identical machine with same cost. Revenue is unaffected by the age of the machine. Ignoring inflation and tax, determine the optimum replacement cycle. PV factors of the cost of capital of 15% for the respective four years are 0.8696, 0.7561, 0.6575 and 0.5718.

## Solution — ICAI

**26. Working Notes**

First of all, we shall calculate cash flows for each replacement cycle as follows:

**One Year Replacement Cycle** ₹

| Year | Replacement Cost | Maintenance & Repair | Residual Value | Net cash Flow |
|---|---|---|---|---|
| 0 | (60,000) | - | - | (60,000) |
| 1 | - | (16,000) | 32,000 | 16,000 |

**Two Years Replacement Cycle** ₹

| Year | Replacement Cost | Maintenance & Repair | Residual Value | Net cash Flow |
|---|---|---|---|---|
| 0 | (60,000) | - | - | (60,000) |
| 1 | - | (16,000) | - | (16,000) |
| 2 | - | (22,000) | 24,000 | 2,000 |

**Three Years Replacement Cycle** ₹

| Year | Replacement Cost | Maintenance & Repair | Residual Value | Net cash Flow |
|---|---|---|---|---|
| 0 | (60,000) | - | - | (60,000) |
| 1 | - | (16,000) | - | (16,000) |
| 2 | - | (22,000) | - | (22,000) |
| 3 | - | (28,000) | 16,000 | (12,000) |

**Four Years Replacement Cycle** ₹

| Year | Replacement Cost | Maintenance & Repair | Residual Value | Net cash Flow |
|---|---|---|---|---|
| 0 | (60,000) | - | - | (60,000) |
| 1 | - | (16,000) | - | (16,000) |
| 2 | - | (22,000) | - | (22,000) |
| 3 | - | (28,000) | - | (28,000) |
| 4 | - | (36,000) | 8,000 | (28,000) |

Now we shall calculate NPV for each replacement cycles

| | | 1 Year | | 2 Years | | 3 Years | | 4 Years | |
|---|---|---|---|---|---|---|---|---|---|
| Year | PVF@ 15% | Cash Flows | PV | Cash Flows | PV | Cash Flows | PV | Cash Flows | PV |
| 0 | 1 | -60,000 | -60,000 | -60,000 | -60,000 | -60,000 | -60,000 | -60,000 | -60,000 |
| 1 | 0.8696 | 16,000 | 13,914 | -16,000 | -13,914 | -16,000 | -13,914 | -16,000 | -13,914 |
| 2 | 0.7561 | - | - | 2,000 | 1,512 | -22,000 | -16,634 | -22,000 | -16,634 |
| 3 | 0.6575 | - | - | - | 0 | -12,000 | -7,890 | -28,000 | -18,410 |
| 4 | 0.5718 | - | - | - | 0 | | 0 | -28,000 | -16,010 |
| | | | -46,086 | | -72,402 | | -98,438 | | -1,24,968 |

| Replacement Cycles | | EAC (₹) |
|---|---|---|
| 1 Year | 46,086 / 0.8696 | 52,997 |
| 2 Years | 72,402 / 1.6257 | 44,536 |
| 3 Years | 98,438 / 2.2832 | 43,114 |
| 4 Years | 1,24,968 / 2.855 | 43,772 |

Since EAC is least in case of replacement cycle of 3 years hence machine should be replaced after every three years.

## Practical Q27

27. Trouble Free Solutions (TFS) is an authorized service center of a reputed domestic air conditioner manufacturing company. All complaints/service related matters of Air conditioner are attended by this service center. The service center employs a large number of mechanics, each of whom is provided with a motor bike to attend the complaints. Each mechanic travels approximately 40000 kms per annuam. TFS decides to continue its present policy of always buying a new bike for its mechanics but wonders whether the present policy of replacing the bike every three year is optimal or not. It is of believe that as new models are entering into market on yearly basis, it wishes to consider whether a replacement of either one year or two years would be better option than present three year period. The fleet of bike is due for replacement shortly in near future.

The purchase price of latest model bike is ₹ 55,000. Resale value of used bike at current prices in market is as follows:

| Period | ₹ |
|---|---|
| 1 Year old | 35,000 |
| 2 Year old | 21,000 |
| 3 Year old | 9,000 |

Running and Maintenance expenses (excluding depreciation) are as follows:

| Year | Road Taxes Insurance etc. (₹) | Petrol Repair Maintenance etc. (₹) |
|---|---|---|
| 1 | 3,000 | 30,000 |
| 2 | 3,000 | 35,000 |
| 3 | 3,000 | 43,000 |

Using opportunity cost of capital as 10% you are required to determine optimal replacement period of bike.

## Solution — ICAI

**27.** In this question the effect of increasing running cost and decreasing resale value have to be weighted upto against the purchase cost of bike. For this purpose, we shall compute Equivalent Annual Cost (EAC) of replacement in different years shall be computed and compared.

| Year | Road Taxes (₹) | Petrol etc. (₹) | Total (₹) | PVF @10% | PV (₹) | Cumulative PV (₹) | PV of Resale Price (₹) | Net Outflow (₹) |
|---|---|---|---|---|---|---|---|---|
| 1 | 3,000 | 30,000 | 33,000 | 0.909 | 29,997 | 29,997 | 31,815 | (1,818) |
| 2 | 3,000 | 35,000 | 38,000 | 0.826 | 31,388 | 61,385 | 17,346 | 44,039 |
| 3 | 3,000 | 43,000 | 46,000 | 0.751 | 34,546 | 95,931 | 6,759 | 89,172 |

**Computation of EACs**

| Year* | Purchase Price of Bike (₹) | Net Outflow (₹) | Total Outflow (₹) | PVAF @ 10% | EAC♣ (₹) |
|---|---|---|---|---|---|
| 1 | 55,000 | (1,818) | 53,182 | 0.909 | 58,506 |
| 2 | 55,000 | 44,039 | 99,039 | 1.735 | 57,083 |
| 3 | 55,000 | 89,172 | 1,44,172 | 2.486 | 57,993 |

Thus, from above table it is clear that EAC is least in case of 2 years, hence bike should be replaced every two years.

* Assume these periods are the periods from which bike shall be kept in use.

♣ EAC is used to bring Cash Flows occurring for different periods at one point of Time.

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